MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, READ ONLY, PROGRAMMABLE, DIGITAL, KEY CODE GYC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

* 1. SCOPE

1.1 SCOPE. This document establishes the requirements and test procedures for the acquisition of an electronic digital programmable read only memory module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

* 2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

* 3.2 Application requirements. This specification establishes the general requirements and test procedures for a programmable read only memory (PROM) module. When A Standard Electronic Modules Program (SEMP) user determines a requirement for a PROM module, the user shall prepare a design approval request (DAR) and submit the DAR to the Standard Electronic Modules Program Design Review Activity (SEMP-DRA) in accordance with MIL-STD-1378. The module user shall prepare a separate detailed module control document.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.2.1 Design Approval Request (DAR) requirement. When a user determines a requirement for a GYC module, the user shall prepare a DAR and submit the DAR to SEMP-DRA in accordance with MIL-STD-1378. The purpose of the DAR requirements is to notify the SEMP-DRA of additional GYC module users and enable the SEMP-DRA to assign GYC dash numbers in accordance with 3.2.2.1. The user shall include in the DAR the estimated number of GYC dash numbers required and the user's detailed module document number.

3.2.2 Detailed module document. The GYC module user shall prepare a separate detailed module document. This document shall carry a system drawing number and all system parts lists and the module shall reference the user's detailed module document number and not M28787/168. The detailed module document shall specify, as a minimum, the module key code as specified in 3.2.2.1, detailed module document number marking, firmware pattern (truth table), module programming technique (method) and marking as specified in 3.2.2.3, detailed electrical test requirements for a programmed module, and MIL-M-28787/168 requirements.

* 3.2.2.1 Module key code assignment. The module user is free to select a marking scheme but the module marking scheme (key code) shall not be a three letter key code other than GYC or a letter-number-letter key code which has been reserved for read only memory (ROM) modules by the SEMP-DRA. The SEMP-DRA has developed a controlled marking scheme which utilizes the GYC key code. Upon request by a DAR form the SEMP-DRA will assign a block of dash numbers which the user may mark on the module fin to distinguish between the different GYC programmed modules. This block of dash numbers will not be issued to other users even if all the dash numbers assigned are not used. For example, company A estimates a need for 75 unique GYC modules. Company A completes a DAR and requests a block of 75 GYC-dash numbers. The SEMP-DRA would assign GYC-01200 through GYC-01275 to company A. This assignment indicates the SEMP-DRA had already assigned GYC-00001 through GYC-01199 to other users. Company A would then mark GYC-01200 to the first pattern requirement and continue until GYC-01275 was used. If additional dash numbers are required, the user shall request them from the SEMP-DRA. NOTE. Under no circumstances shall a user assign a key code or dash numbers.

3.2.2.2 Firmware pattern. The module firmware pattern (truth table) shall be specified in the detailed module document. If the key code-dash number scheme of 3.2.2.1 is used, the different firmware patterns are specified by dash numbers. The different firmware patterns could be specified as a part number to the detailed module document. For example, if the detailed document number was 2658555, the first firmware pattern would be specified as 2658555P1. The nth firmware pattern would be specified as 2658555Pn.

3.2.2.3 Program technique and marking. The detailed programming technique (method) shall be specified in the detailed module document. Because of the numerous unique programming methods, the user must determine the number of unique programming methods allowed. The detailed module document shall specify the allowed programming procedure and the corresponding module marking procedure. As a minimum, the detailed module document shall specify: The exact step-by-step programming procedure for each programming option, minimum and maximum programming electrical parameters, maximum number of programming attempts to program a selected bit, programming time and duty cycle, maximum module temperature, and the exact module voltages and output load current to perform program verifications. For example, company A has surveyed the potential module vendors and programming techniques and has elected to allow two programming methods. Because of potential programming technique changes, company A has decided to specify each programming method by two separate programming documents. The detailed module document specifies that the module shall be programmed in accordance with the selected program method and the module shall be marked with the corresponding program document number including revision level as applicable.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787. All unprogrammed modules shall be tested on a correlated test system. All programmed modules shall be tested as specified in the user's detailed module document.

* 3.4 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.727 failures per million operating hours when predicted in accordance with MIL-STD-1389.

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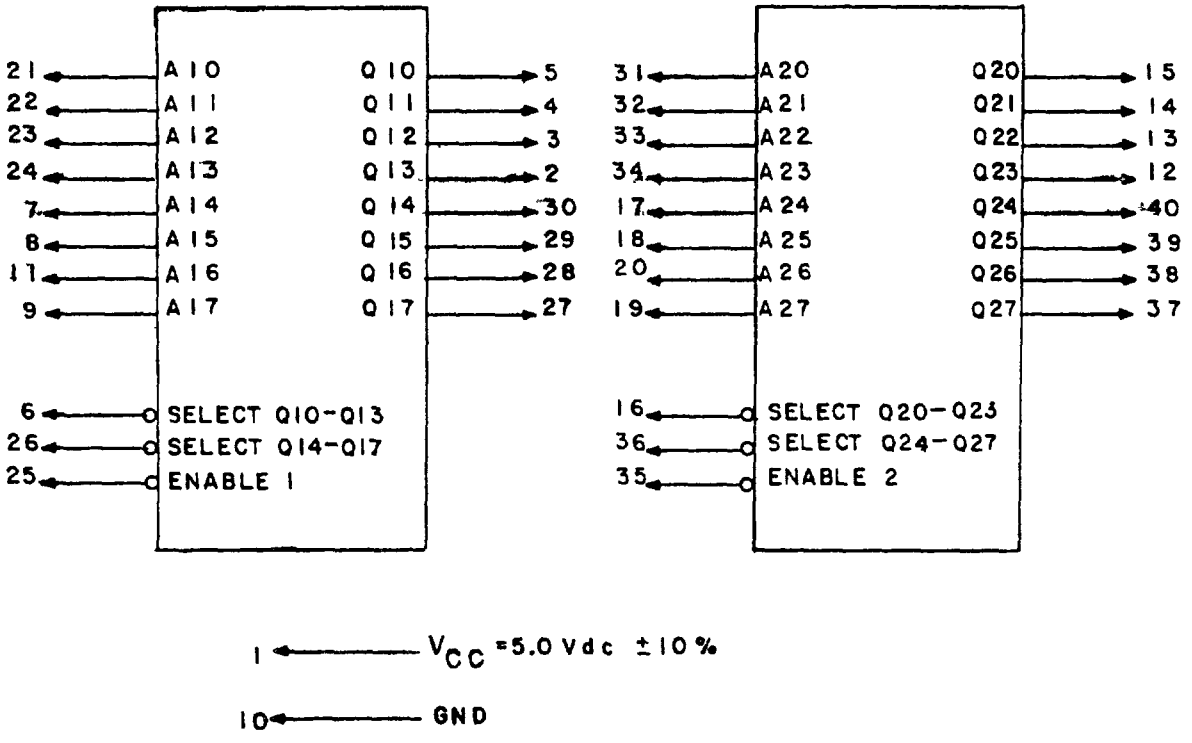


FIGURE 1 GYC module function diagram, programmable read only memory (PROM), dual 256-word by 8-bit.

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TABLE I. Module electrical requirements.

	Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
					Initial		End-of-life		Initial		End-of-life		
					Min	Max	Min	Max	Min	Max	Min	Max	
	A	Isolation	II	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
	B1-B18	Input breakdown current, address and enable, V_{IH} = 5.5 V	III	S	0	2	0	2	0	2	0	2	mA
	B19-B22	Input breakdown current, select, V_{IH} = 4.5 V	III	S	-0.1	1	-0.1	1	-0.1	1	-0.1	1	mA
	C1-C16	Output short circuit current	IV	S	-15	-90	-15	-90	-15	-90	-15	-90	mA
	D1-D16	V_{OUT} (low-loaded)	V	S	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V
	E1-E22	Input clamp diode voltage	VI	S	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	V
*	F1-F18	Input current (address and enable), low level	VII	X	0	-0.56	0	-0.56	0	-0.56	0	-0.56	mA
*	F19-F22	Input current (select), low level	VII	X	0	-0.28	0	-0.28	0	-0.28	0	-0.28	mA
	G1-G18	Input current (address and enable), high level	VII	X	-20	80	-20	80	-20	80	-20	80	µA
	G19-G22	Input current (select), high level	VII	X	-20	40	-20	40	-20	40	-20	40	µA
	H	Power supply current	VIII	X	10	450	10	450	10	450	10	450	mA
	J1-J16	Output leakage current (high-impedance state), V_O = 2.4 V	IX	X	-100	100	-100	100	-100	100	-100	100	µA
	J17-J32	Output leakage current (high-impedance state), V_O = 0.45 V	IX	X	-100	100	-100	100	-100	100	-100	100	µA
	K	Output voltage (V_O) level (unprogrammed)	X	X	← See table X, no failure →								
	L	Power supply filter capacitance	XI	X	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
	M	Programmability	XII	Q	← See table XII, no failure →								
	N	Functional test pattern	XIV	Q	← See table XIV, no failure →								
	P1-P16	Static output voltage, low level	XV	Q	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	V
	Q1-Q16	Static output voltage, high level	XV	Q	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
	R1-R16	Transition time (t_{TLH}), C_L = 50 pF	XVI	Q	5	30	5	30	5	30	5	30	ns

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-4/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R17-R32	Transition time (t_{TLH}), C_L = 150 pF	XVI	Q	5	40	5	40	5	40	5	40	ns
* S1-S16	Transition time (t_{THL}), C_L = 50 pF	XVI	Q	3	30	3	30	3	30	3	30	ns
S17-S32	Transition time (t_{THL}), C_L = 150 pF	XVI	Q	5	40	5	40	5	40	5	40	ns
T1-T32	Enable/select recovery time (t_{ERL}), C_L < 10 pF	XVII	Q	5	40	5	40	5	40	5	40	ns
U1-U32	Enable/select recovery time (t_{ERH}), C_L < 10 pF	XVII	Q	5	40	5	40	5	40	5	40	ns
V1	Galloping read access pattern (C_L = 50 pF, V_{CC} = 4.5 V)	XVIII	Q	← See table XVIII, no failure →								
V2	Galloping read access pattern (C_L = 50 pF, V_{CC} = 5.5 V)	XVIII	Q	← See table XVIII, no failure →								
V3	Galloping read access pattern (C_L = 150 pF, V_{CC} = 4.5 V)	XVIII	Q	← See table XVIII, no failure →								
V4	Galloping read access pattern (C_L = 150 pF, V_{CC} = 5.5 V)	XVIII	Q	← See table XVIII, no failure →								
W1-W6	Input crosstalk (enable and select), low level	XIX	Q	0	400	0	400	0	400	0	400	mV
X1-X6	Input crosstalk (enable and select), high level	XIX	Q	0	400	0	400	0	400	0	400	mV
Y	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
Z1-Z18	Input capacitance (address and enable)	XXI	Q	2	30	2	30	2	30	2	30	pF
Z19-Z22	Input capacitance (select)	XXI	Q	2	20	2	20	2	20	2	20	pF
Z23-Z38	Output capacitance	XXI	Q	2	15	2	15	2	15	2	15	pF

1/ See order of testing (4.4.2).
2/ "X" parameters shall be 100 percent tested during module acceptance tests.
3/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
4/ "Q" parameters shall be tested during qualification testing only.

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3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I, the user's detailed module document (see 3.2.2.), and shall operate within the environment of design class specified.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389, the user's detailed module document (see 3.2.2), and as specified herein.

3.6.1 Key code. The module key code shall be as specified in the user's detailed module document.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and in the user's detailed module document, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.8 Unprogrammed GYC modules. Unprogrammed modules shall be as specified in the user's detailed module document and as specified herein.

3.8.1 Unprogrammed output state. All unprogrammed module outputs shall be in the same output state (high or low level) and the state shall be as specified in the user's detailed module document.

3.8.2 Module programming. All modules shall be capable of being programmed using the module connector pins and using the appropriate programming technique as specified in the user's detailed module document.

3.8.3 Module programming technique marking. The programming technique necessary to program an individual module shall be marked on that individual module by the module vendor in accordance with the user's detailed module document.

3.9 Freeze-out test. The freeze-out test shall be performed on all PROM integrated circuits if the fuse is formed with a thin film metallic (including nichrome) material. The test shall be perfomred in accordance with 4.6. The freeze-out test shall be performed on all devices selected for group B testing of method 5005 for class B of MIL-STD-883 and one failure shall constitute failure of the lot.

3.10 Acceptance test. All production modules shall be tested in accordance with notes 2 and 3 of table I and as specified in the user's detailed module document.

3.11 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2)

* 3.12 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 $\pm$0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc $\pm$1 percent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787 and in the user's detailed module document.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

4.4.2 Order of testing. The order of testing for unprogrammed and programmed modules shall be as follows:

4.4.2.1 Unprogrammed module order of testing. Test conditions A through L shall be performed on all unprogrammed modules. Test condition order shall be A, B, C, and D followed by E through L in any order.

4.4.2.2 Programmed module order of testing. Test conditions A through J and L through Z shall be performed on all programmed modules. Test condition order shall be M, A, B, C, and D followed by E through J, L and N through Z in any order.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modification or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through L in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient on all unprogrammed modules. Test conditions A through J, L and N through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient on all programmed modules. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through L, and N through Y, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column, and test conditions N, P, and Q, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test, electrical tests E through Y, shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

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4.6 Freeze-out test. Within no more than 24 hours after completion of burn-in, or if more than 24 hours have elapsed subsequent to the 125°C burn-in exposure, devices shall be conditioned with at least 125° for a minimum of five hours immediately prior to the freeze-out test. The freeze-out test shall be conducted as follows:

a. Connect devices in the electrical configuration of test condition C, method 1015 of MIL-STD-883, except that the bias shall be cycled, 3 minutes "on" and 3 minutes "off", throughout the duration of the test.
b. Reduce the device temperature to a maximum TA of -10°C and maintain them at that temperature for a minimum of 5 hours duration.
c. With the cycle bias maintained, allow TA to go to room temperture (by removal from the cold chamber or termination of forced cooling but with no forced heating or forced-air flow) and retain for a minimum of 19 hours subsequent to the completion of the 5-nour cold soak. TA shall not exceed 35°C during this period.
d. Remove bias and subject devices to the final electrical tests as specified.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.11).

* 6.3 Cross reference of classification. Part number M28787/168-1 supersedes part number M28787/168. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	2.5 W (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, address, enable	
Low level (V_{IL} = 0.0 V)	-3.46 mA maximum
(V_{IL} = 0.4 V)	-3.2 mA maximum
High level (V_{IH} = 2.4 V)	80 µA maximum
(V_{IH} = 5.5 V)	2 mA maximum
Input current, select (program pin)	
Low level (V_{IL} = 0.0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.6 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 4.5 V)	1 mA maximum
Input clamp diode voltage	
I_{clamp} = -5 mA	-0.2 V minimum, -1.5 V maximum
Output voltage	
Low level (I_{OL} = 10 mA)	0.45 V maximum
High level (I_{OH} = -900 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.45 V)	10 mA
High level (V_{OH} = 2.4 V)	-900 µA

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Output short circuit current	
V_{OL} = 0.0 V, V_{CC} = 5.5 V	-15 mA minimum, -90 mA maximum
Output leakage current (high-impedance state)	
0.45 ≤ Vo ≤ 2.4 V	±100 µA maximum
Propagation delay time (see figure 2)	
t_{AA} - Address to data output	
C_L = 50 pF	80 ns maximum
C_L = 150 pF	90 ns maximum
t_{EA} - Enable/select	
C_L = 50 pF	45 ns maximum
C_L = 150 pF	55 ns maximum
Enable/select recovery time	
t_{ERL} - Low-state to high-impedance state (C_L < 10 pF)	40 ns maximum
t_{ERH} - High-state to high-impedance state (C_L < 10 pF)	40 ns maximum

* 6.5 Module function. The programmable read only memory (PROM) module is organized as two independent 256-word by 8-bit programmable read only memories. These memories may be interconnected through connector wiring to form a memory of either 256 by 16 or 512 by 8.

* 6.5.1 Input to output relationship. The module contains two independent 256-word by 8-bit programmable read only memories (PROM) as shown on figure 1. When the enable and select of one of the 256-word by 8-bit memories goes to the low level, the data programmed at the selected address input appears at the eight data outputs of that memory. When the enable or the select of one of the memories is high, the eight output lines of that memory are disabled to a high-impedance state. Module timing and truth table are defined on figure 2.

* 6.5.2 Paralleling inputs and outputs. Corresponding address inputs of the two address groups must be paralleled for nonindependent operation of the two memory sections. A number of memory outputs can be connected to a common bus. All of the memories except one are placed in the high-impedance state, and the selescted memory is enabled. When outputs of several modules are bussed together, all address lines should be addressed using synchronous clocking.

* 6.5.3 Programming. Module programming technique (see 3.2.2.3) is defined by the GYC module user.

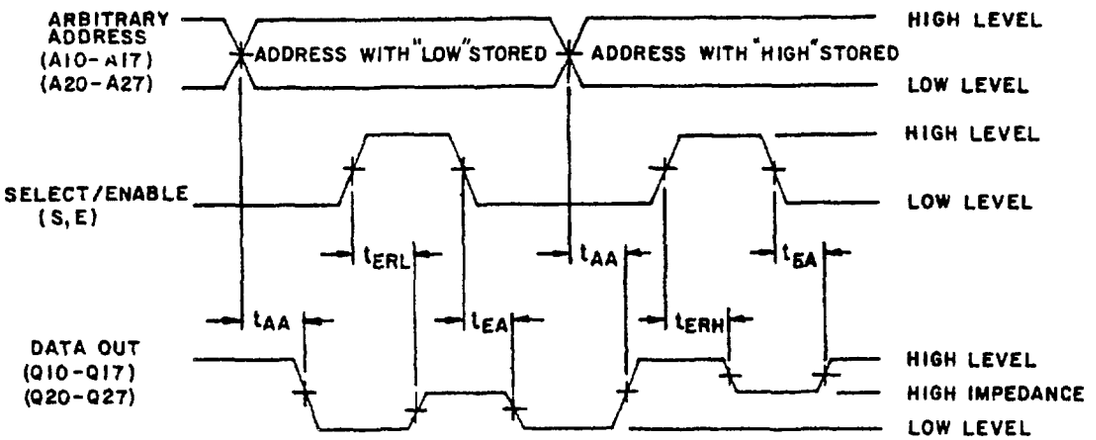
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DYNAMIC CHARACTERISTICS WAVEFORMS

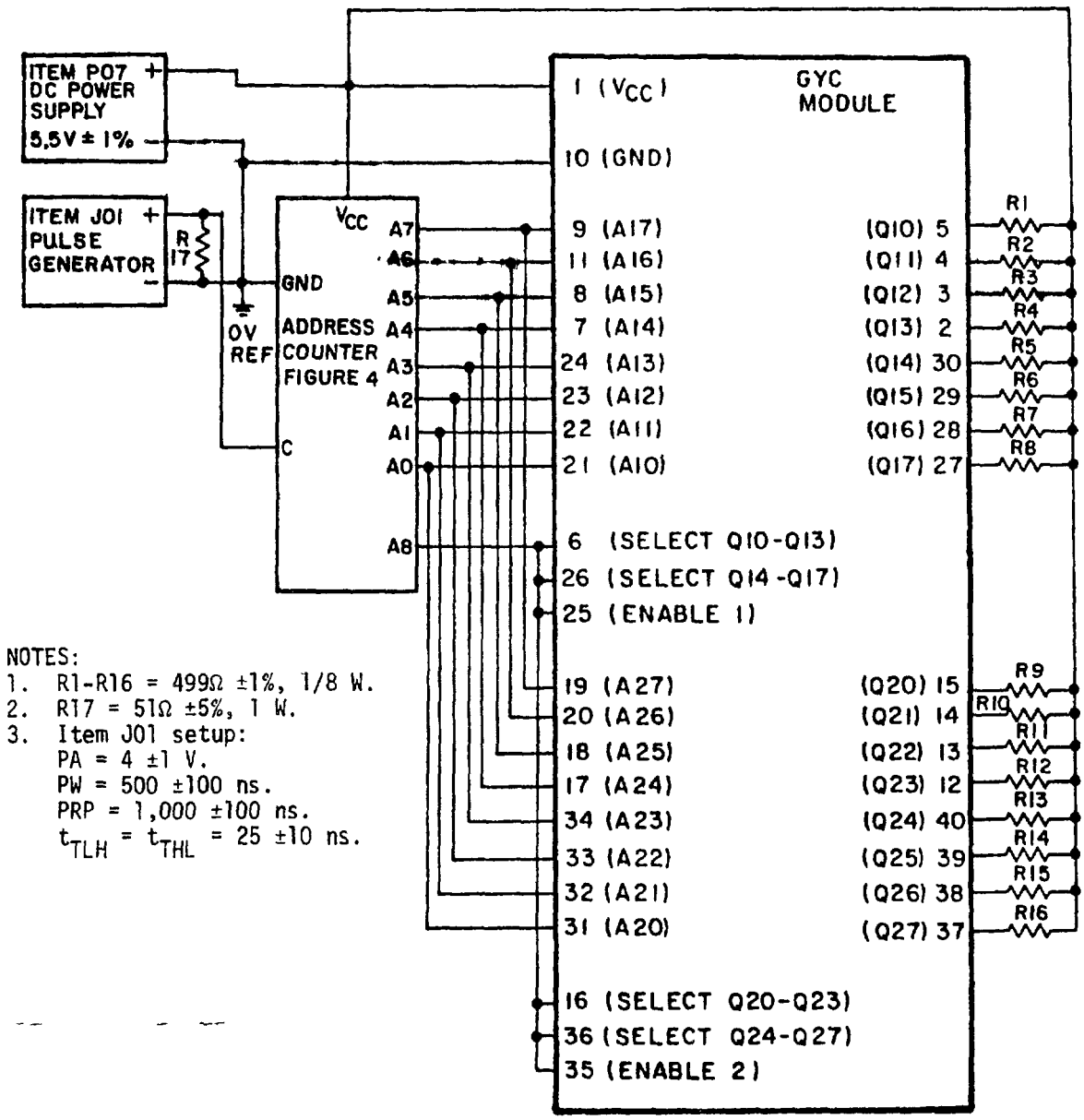
Module truth table.

Select(S)	Select(S)	Enable (E)	Outputs	
Q10-Q13 (Q20-Q23)	Q14-Q17 (Q24-Q27)	1 (2)	Q10-Q13 (Q20-Q23)	Q14-Q17 (Q24-Q27)
X	X	H	High Z	High Z
H	H	L	High Z	High Z
H	L	L	High Z	Read
L	H	L	Read	High Z
L	L	L	Read	Read

NOTES:
1. H = High-level input.
2. L = Low-level input.
3. High Z = High-impedance state.

* FIGURE 2. Dynamic characteristics waveforms and module truth table.

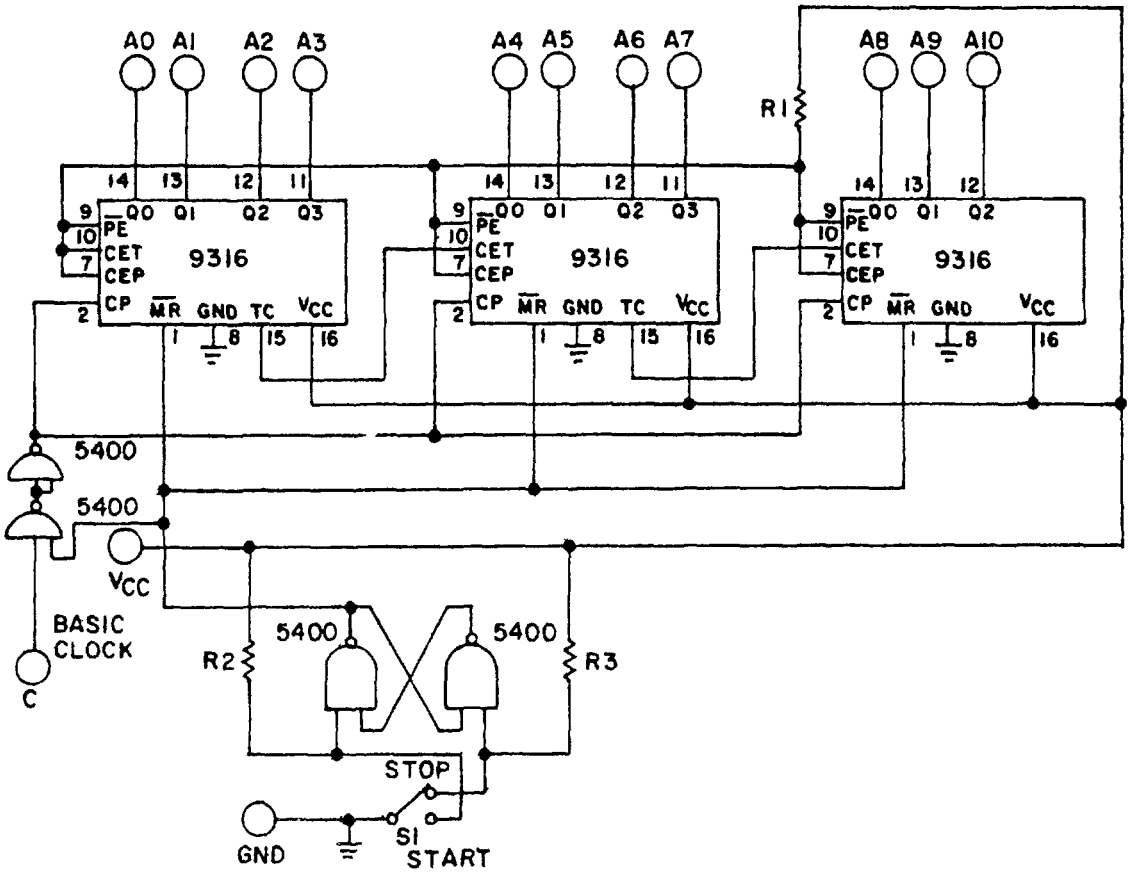
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NOTES:
1. R1-R16 = 499Ω ±1%, 1/8 W.
2. R17 = 51Ω ±5%, 1 W.
3. Item J01 setup:
   PA = 4 ±1 V.
   PW = 500 ±100 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 25 ±10 ns.

FIGURE 3. Operating temperature and life-test circuit.

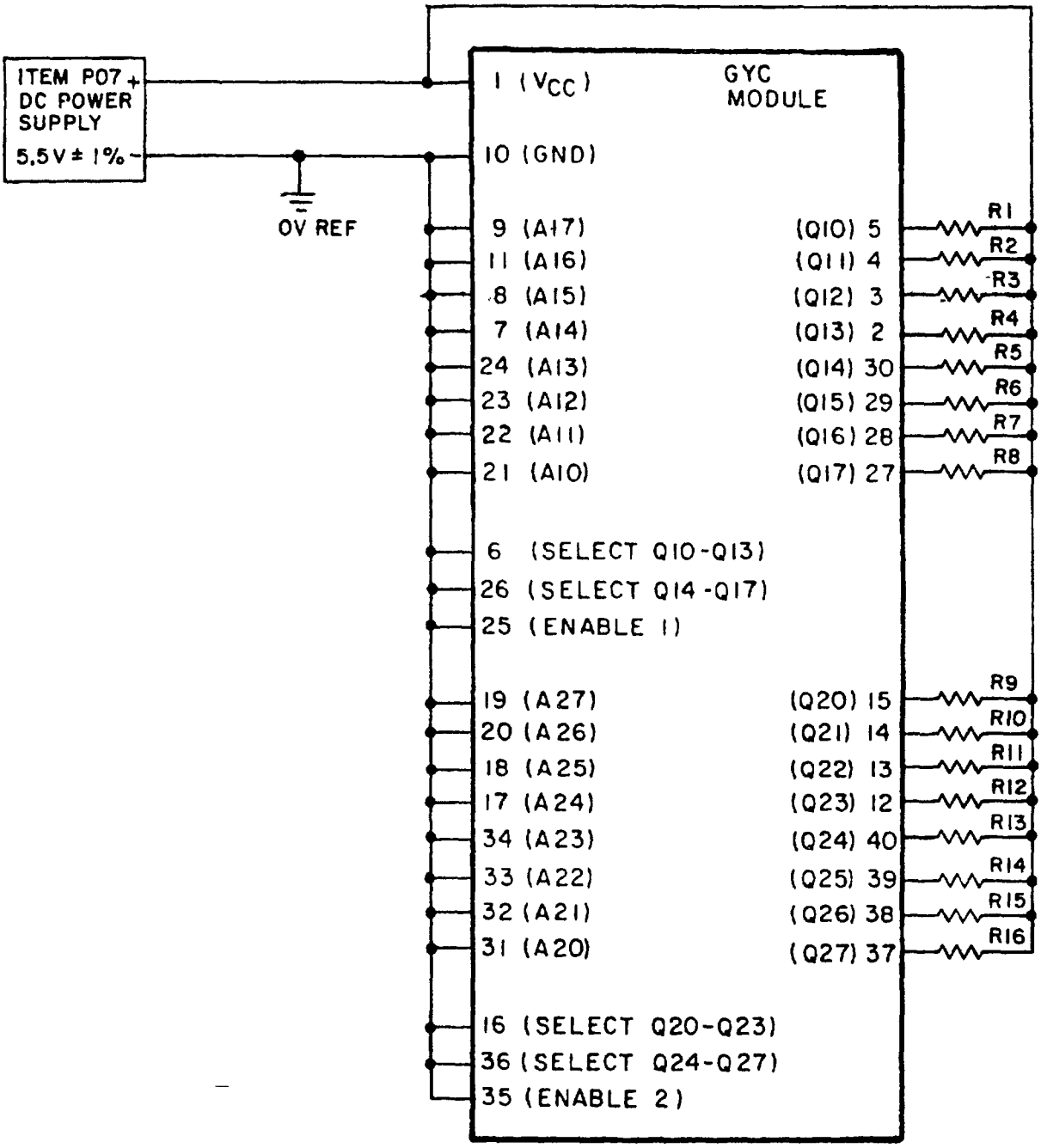
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NOTES:
1. R1-R3 = 2.2 kΩ ±1%, 1/8 W.
2. S1 = SPDT switch.
3. 3 each 9316 high-speed synchronous 4-bit binary counters.
4. 1 each 5400 quad 2-input NAND gate.

FIGURE 4. Address counter circuit.

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NOTE: R1-R16 = 499Ω ±1%, 1/8 W.

FIGURE 5. Component temperature test circuit.

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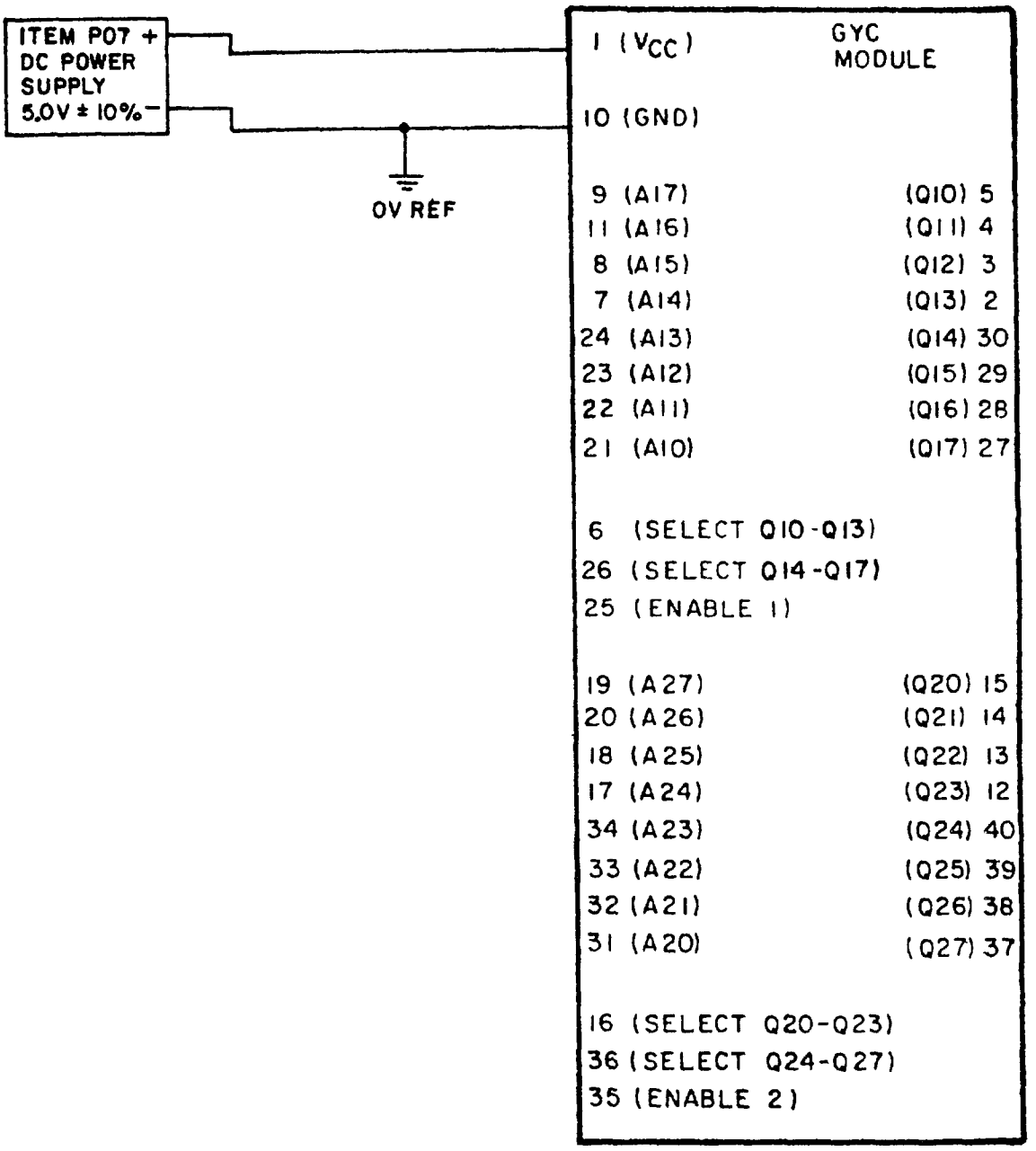


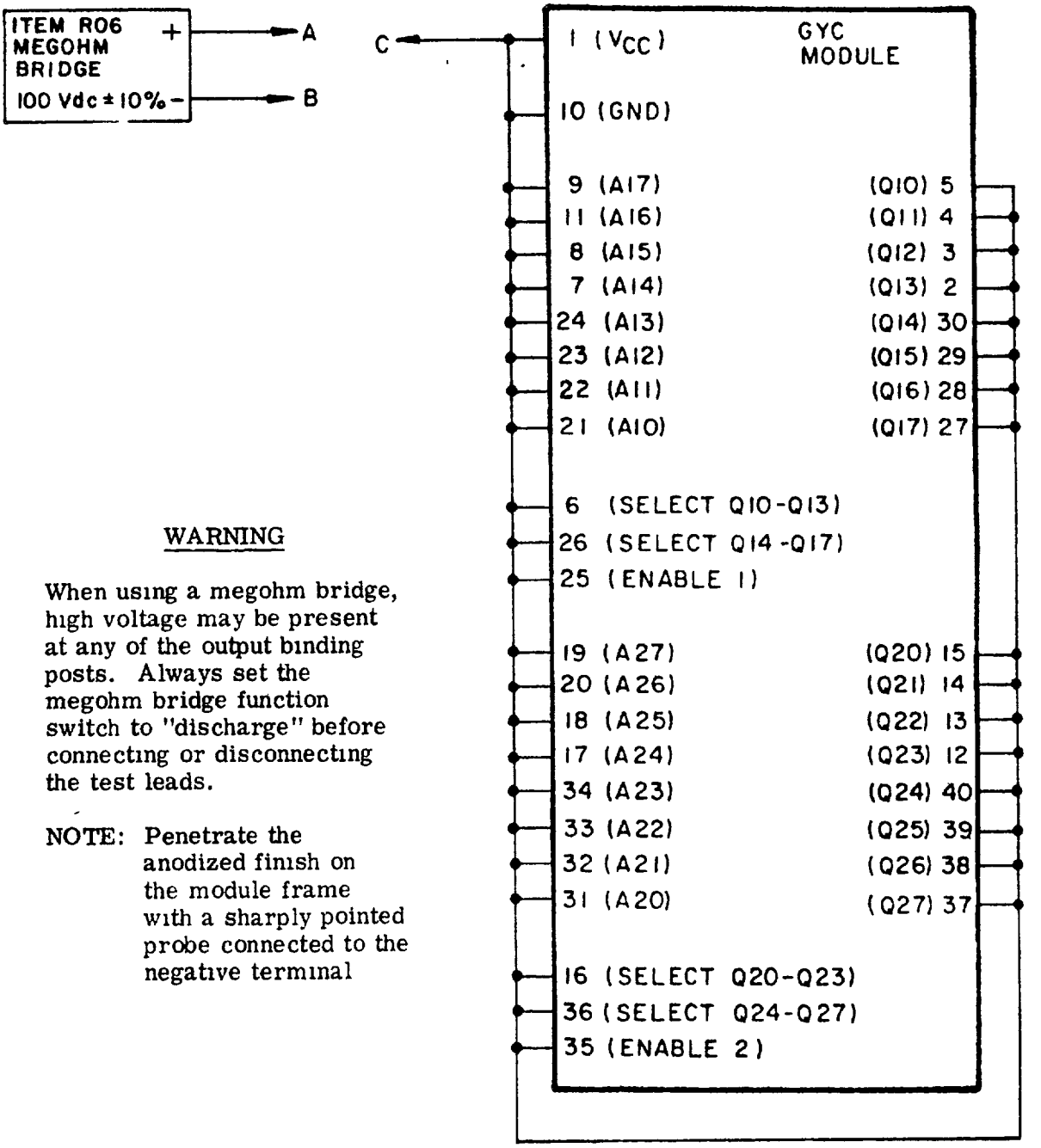
FIGURE 6. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation, frame to module pins	C	Frame	10	∞	MΩ	7

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 7. Isolation test circuit for table II.

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TABLE III Input breakdown current.

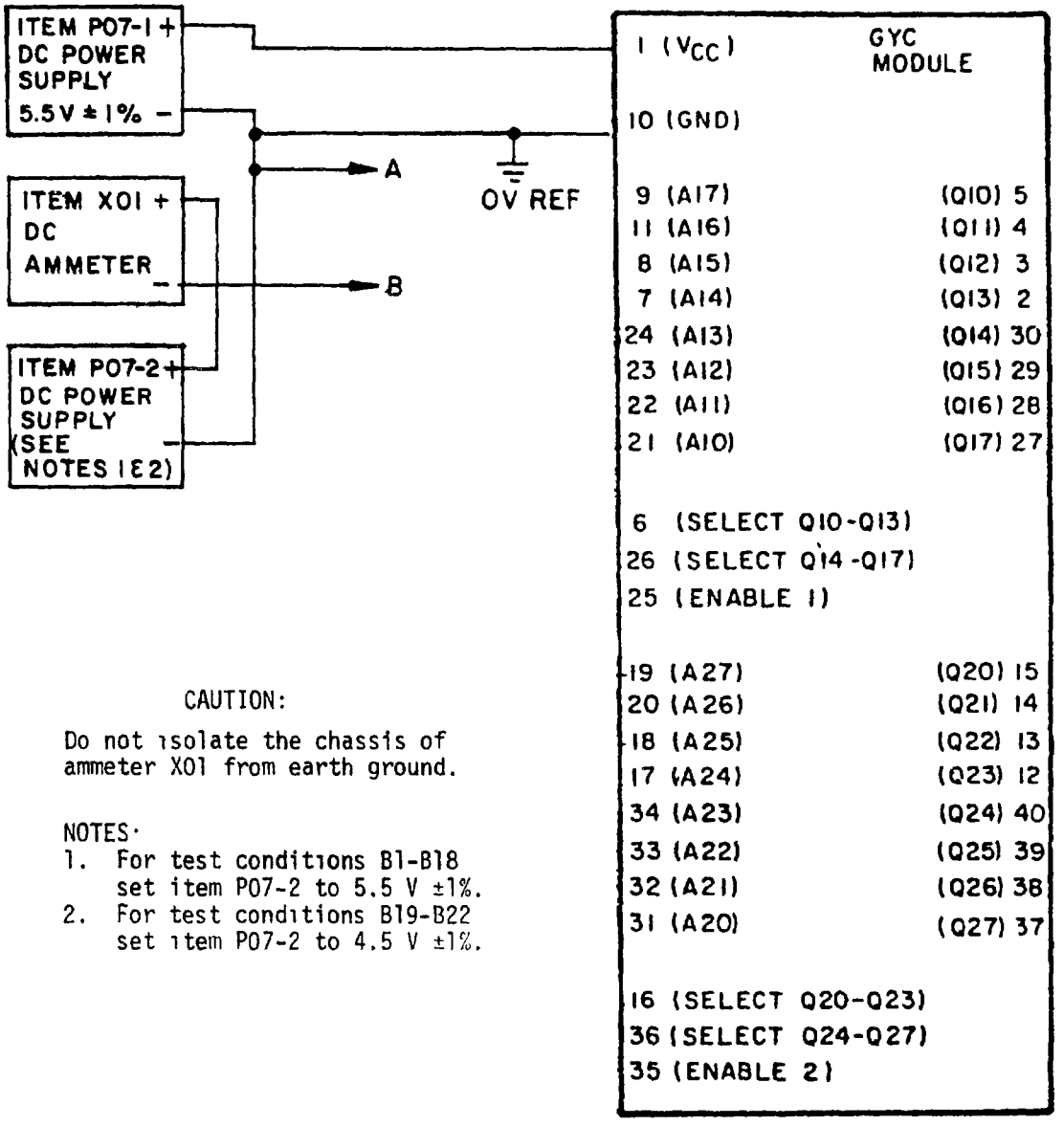
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	A17 Input breakdown current	11,8,7,24,23,22,21,19,20,18,17, 34,33,32,31,25,35,6,26,16,36	9	0	2	mA	8
B2	A16	9,8,7,24,23,22,21,19,20,18,17, 34,33,32,31,25,35,6,26,16,36	11				
B3	A15	9,11,7,24,23,22,21,19,20,18,17, 34,33,32,31,25,35,6,26,16,36	8				
B4	A14	9,11,8,24,23,22,21,19,20,18,17, 34,33,32,31,25,35,6,26,16,36	7				
B5	A13	9,11,8,7,23,22,21,19,20,18,17, 34,33,32,31,25,35,6,26,16,36	24				
B6	A12	9,11,8,7,24,22,21,19,20,18,17, 34,33,32,31,25,35,6,26,16,36	23				
B7	A11	9,11,8,7,24,23,21,19,20,18,17, 34,33,32,31,25,35,6,26,16,36	22				
B8	A10	9,11,8,7,24,23,22,19,20,18,17, 34,33,32,31,25,35,6,26,16,36	21				
B9	A27	9,11,8,7,24,23,22,21,20,18,17, 34,33,32,31,25,35,6,26,16,36	19				
B10	A26	9,11,8,7,24,23,22,21,19,18,17, 34,33,32,31,25,35,6,26,16,36	20				
B11	A25	9,11,8,7,24,23,22,21,19,20,17, 34,33,32,31,25,35,6,26,16,36	18				
B12	A24	9,11,8,7,24,23,22,21,19,20,18, 34,33,32,31,25,35,6,26,16,36	17				
B13	A23	9,11,8,7,24,23,22,21,19,20,18, 17,33,32,31,25,35,6,26,16,36	34				
B14	A22	9,11,8,7,24,23,22,21,19,20,18, 17,34,32,31,25,35,6,26,16,36	33				

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TABLE III. Input breakdown current - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B15	A21 Input breakdown current	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,31,25,35,6,26,16,36	32	0	2	mA	8
B16	A20	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,25,35,6,26,16,36	31				
B17	Enable 1	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,35,6,26,16,36	25				
B18	Enable 2	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,6,26,16,36	35				
B19	Select (Q10-Q13)	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,26,16,36	6	-0.1	1		
B20	Select (Q14-Q17)	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,16,36	26				
B21	Select (Q20-Q23)	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26,36	16				
B22	Select (Q24-Q27)	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26,16	36				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:24Z
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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. For test conditions B1-B18 set item P07-2 to 5.5 V ±1%.
2. For test conditions B19-B22 set item P07-2 to 4.5 V ±1%.

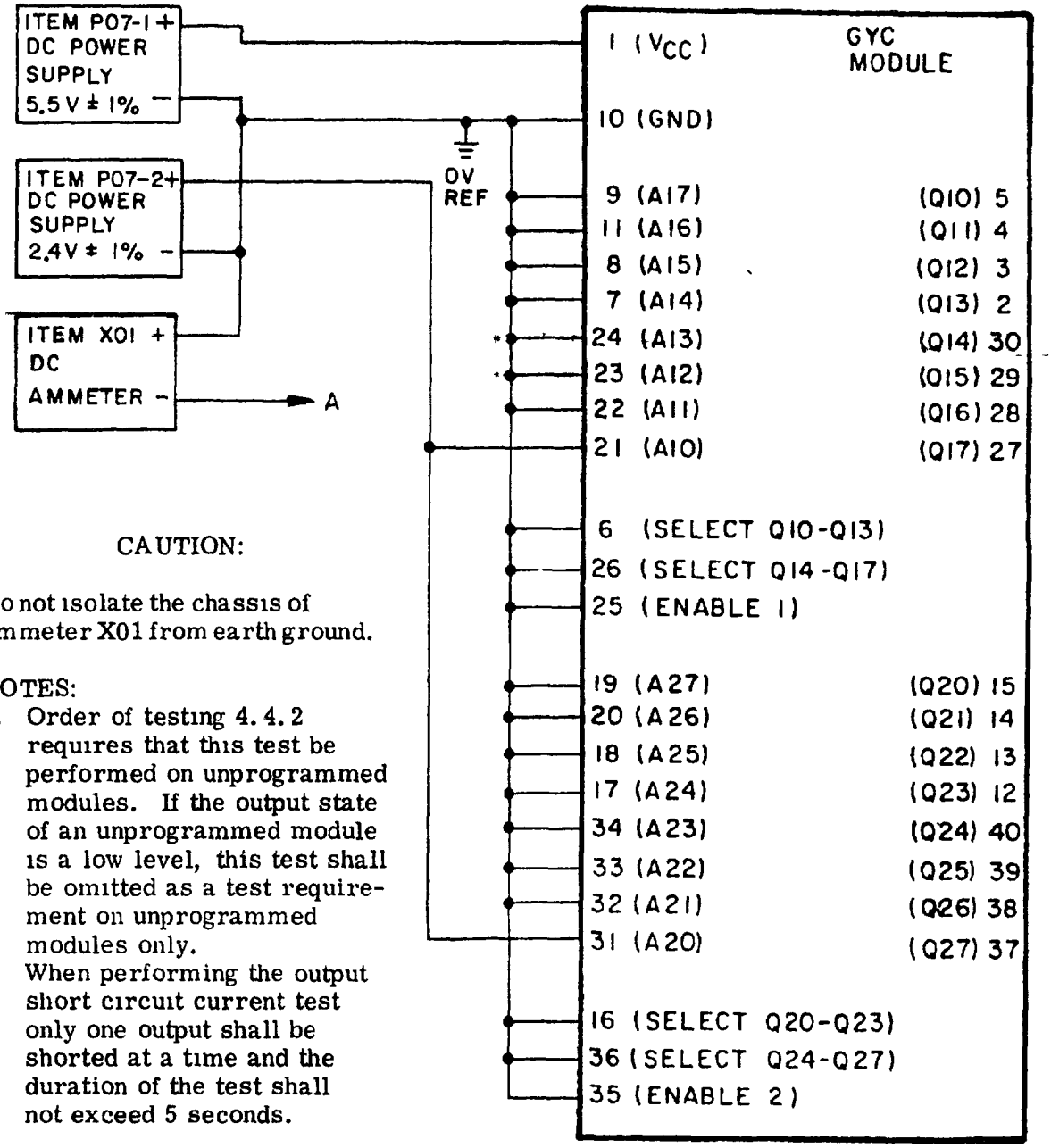
FIGURE 8. Input breakdown current test circuit for table III.

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TABLE IV. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
C1	Q10 Output short circuit current	5	-15	-90	mA	9
C2	Q11	4				
C3	Q12	3				
C4	Q13	2				
C5	Q14	30				
C6	Q15	29				
C7	Q16	28				
C8	Q17	27				
C9	Q24	40				
C10	Q25	39				
C11	Q26	38				
C12	Q27	37				
C13	Q20	15				
C14	Q21	14				
C15	Q22	13				
C16	Q23	12				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:

1. Order of testing 4.4.2 requires that this test be performed on unprogrammed modules. If the output state of an unprogrammed module is a low level, this test shall be omitted as a test requirement on unprogrammed modules only.
2. When performing the output short circuit current test only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

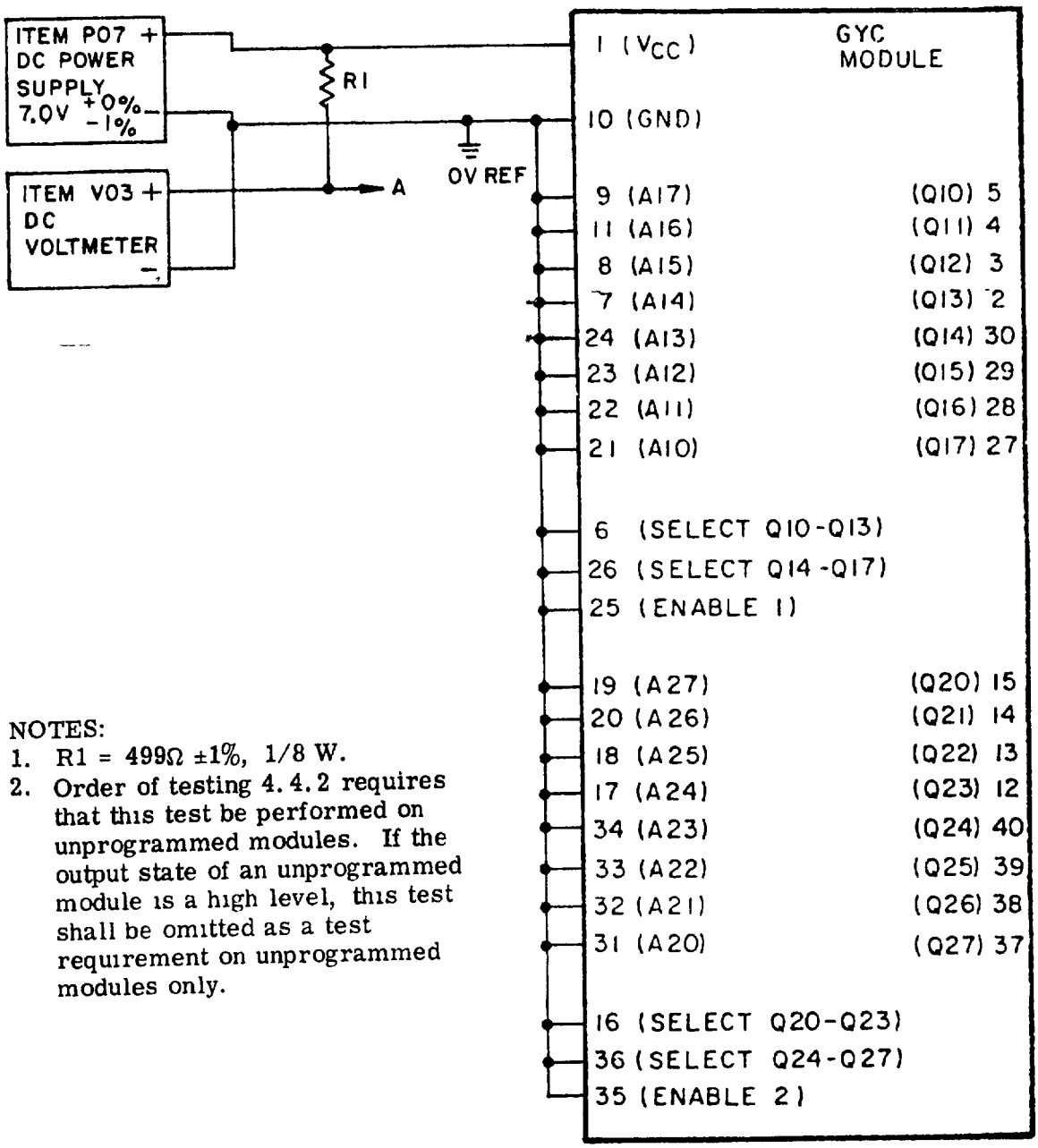
FIGURE 9. Output short circuit current test circuit for table IV.

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TABLE V. Vout (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D1	Q10 Vout (low-loaded)	5	0.0	0.8	V	10
D2	Q11	4				
D3	Q12	3				
D4	Q13	2				
D5	Q14	30				
D6	Q15	29				
D7	Q16	28				
D8	Q17	27				
D9	Q24	40				
D10	Q25	39				
D11	Q26	38				
D12	Q27	37				
D13	Q20	15				
D14	Q21	14				
D15	Q22	13				
D16	Q23	12				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:24Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 499Ω ±1%, 1/8 W.
2. Order of testing 4.4.2 requires that this test be performed on unprogrammed modules. If the output state of an unprogrammed module is a high level, this test shall be omitted as a test requirement on unprogrammed modules only.

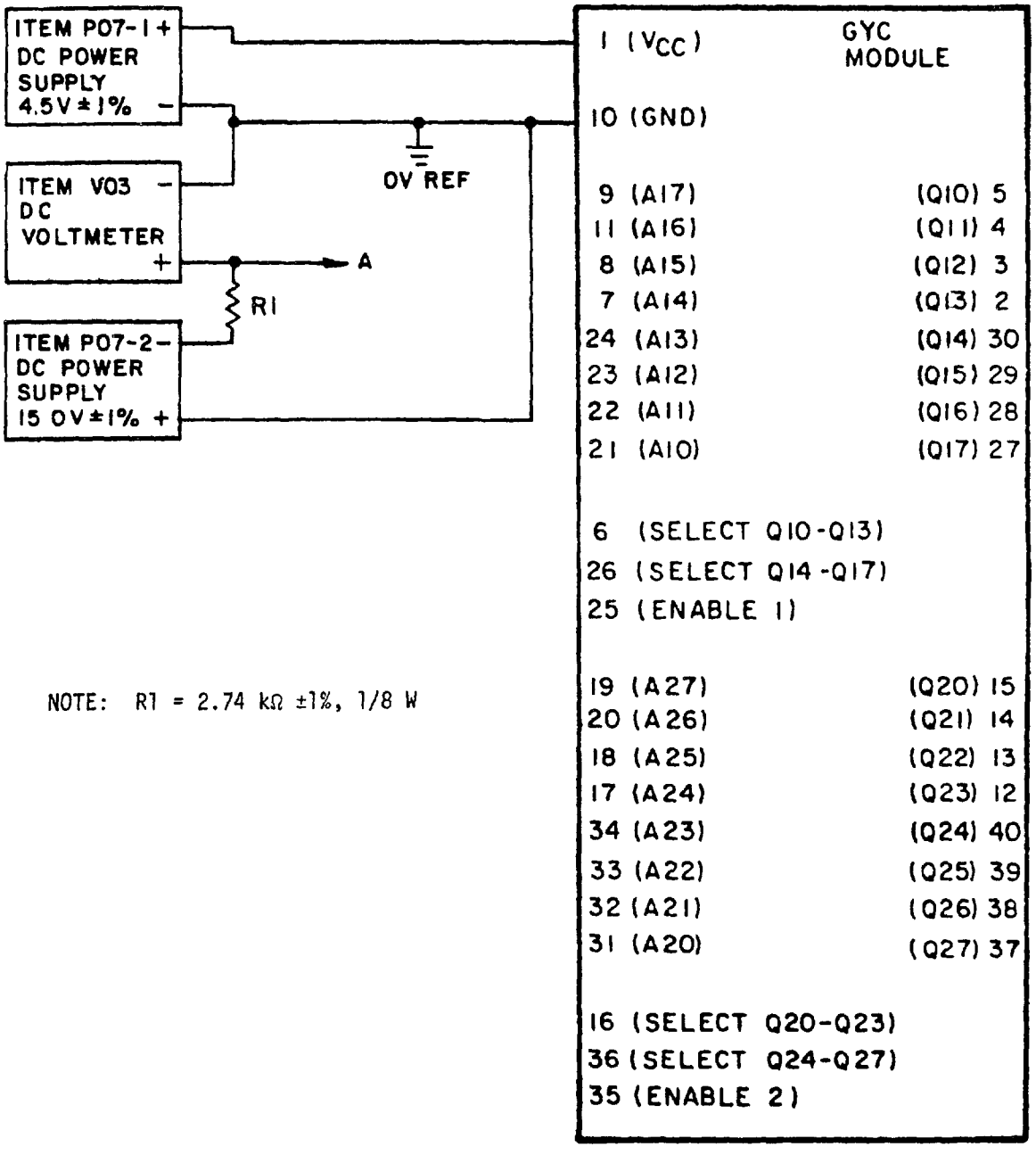
FIGURE 10. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
E1	A17 Input clamp diode voltage	9	-0.2	-1.5	V	11
E2	A16	11				
E3	A15	8				
E4	A14	7				
E5	A13	24				
E6	A12	23				
E7	A11	22				
E8	A10	21				
E9	A27	19				
E10	A26	20				
E11	A25	18				
E12	A24	17				
E13	A23	34				
E14	A22	33				
E15	A21	32				
E16	A20	31				
E17	Select Q10-Q13	6				
E18	Select Q14-Q17	26				
E19	Enable 1	25				
E20	Select Q20-Q23	16				
E21	Select Q24-Q27	36				
E22	Enable 2	35				

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NOTE: R1 = 2.74 kΩ ±1%, 1/8 W

*FIGURE 11. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Input current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	A17 Input current, low level	Open	9	Open	0	-0.56	mA	12
F2	A16		11					
F3	A15		8					
F4	A14		7					
F5	A13		24					
F6	A12		23					
F7	A11		22					
F8	A10		21					
F9	A27		19					
F10	A26		20					
F11	A25		18					
F12	A24		17					
F13	A23		34					
F14	A22		33					
F15	A21		32					
F16	A20		31					
F17	Enable 1		25					
F18	Enable 2		35					
F19	Select Q10-Q13		6			-0.28		
F20	Select Q14-Q17		26					
F21	Select Q20-Q23		16					
F22	Select Q24-Q27		36					

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TABLE VII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	A17	Input current, high level	9	Open	11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36	-20	80	µA	12
G2	A16		11		9,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
G3	A15		8		9,11,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
G4	A14		7		9,11,8,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
G5	A13		24		9,11,8,7,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
G6	A12		23		9,11,8,7,24,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
G7	A11		22		9,11,8,7,24,23,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
G8	A10		21		9,11,8,7,24,23,22,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
G9	A27		19		9,11,8,7,24,23,22,21,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
G10	A26		20		9,11,8,7,24,23,22,21,19,18, 17,34,33,32,31,25,35,6,26, 16,36				

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TABLE VII. Input current (low-level and high-level) test - Continued.

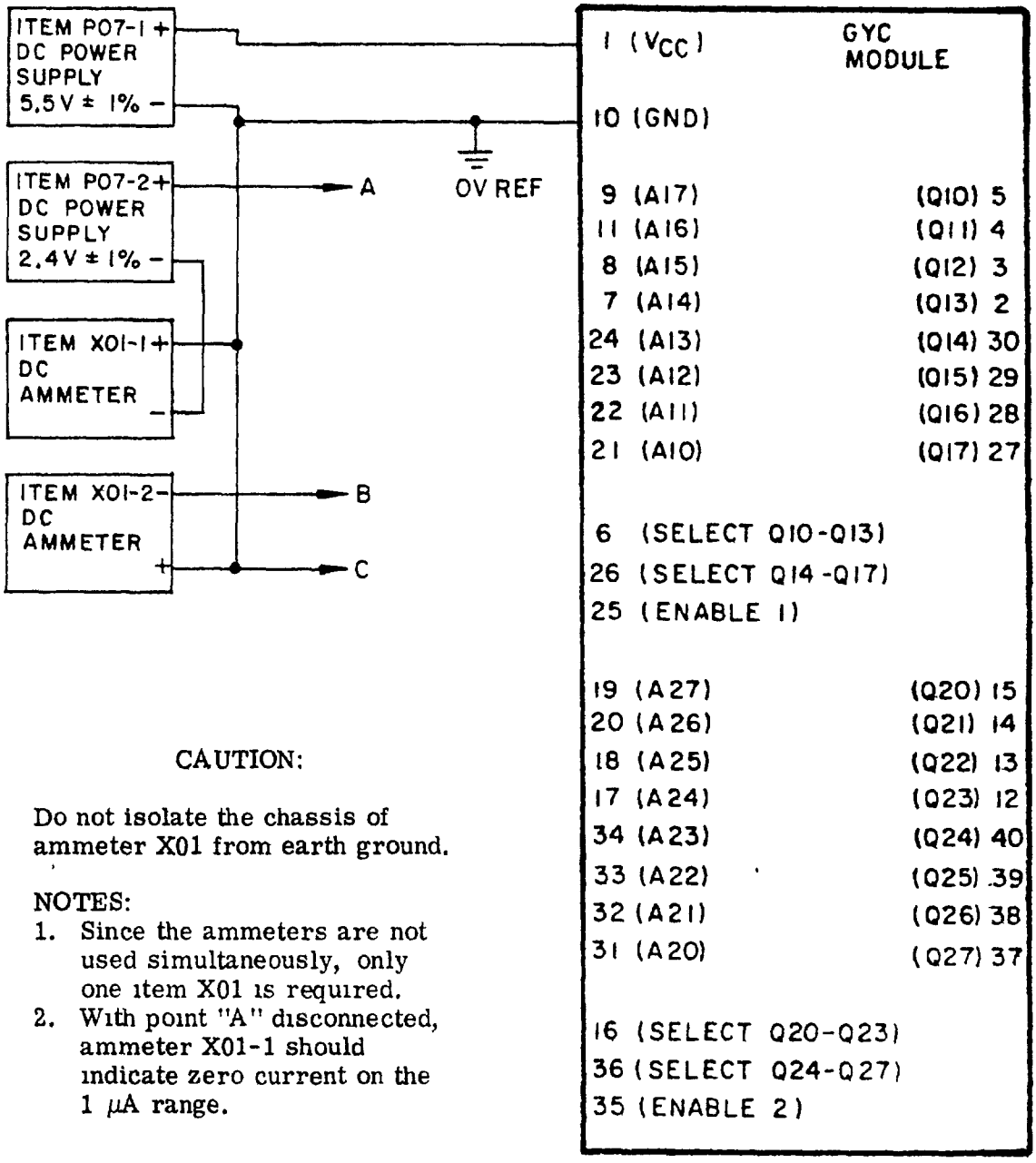
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G11	A25 Input current, high level	18	Open	9,11,8,7,24,23,22,21,19,20, 17,34,33,32,31,25,35,6,26, 16,36	-20	80	µA	12
G12	A24	17		9,11,8,7,24,23,22,21,19,20, 18,34,33,32,31,25,35,6,26, 16,36				
G13	A23	34		9,11,8,7,24,23,22,21,19,20, 18,17,33,32,31,25,35,6,26, 16,36				
G14	A22	33		9,11,8,7,24,23,22,21,19,20, 18,17,34,32,31,25,35,6,26, 16,36				
G15	A21	32		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,31,25,35,6,26, 16,36				
G16	A20	31		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,25,35,6,26, 16,36				
G17	Enable 1	25		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,35,6,26, 16,36				
G18	Enable 2	35		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,25,6,26, 16,36				
G19	Select Q10-Q13	6		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,25,35,26, 16,36		40		
G20	Select Q14-Q17	26		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,25,35,6, 16,36		40		

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TABLE VII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G21	Select Q20-Q23 Input current, high level	16	Open	9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,25,35,6, 26,36	-20	40	µA	12
G22	Select Q24-Q27 Input current, high level	36	Open	9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,25,35,6, 26,16	-20	40	µA	12

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since the ammeters are not used simultaneously, only one item X01 is required.
2. With point "A" disconnected, ammeter X01-1 should indicate zero current on the 1 μA range.

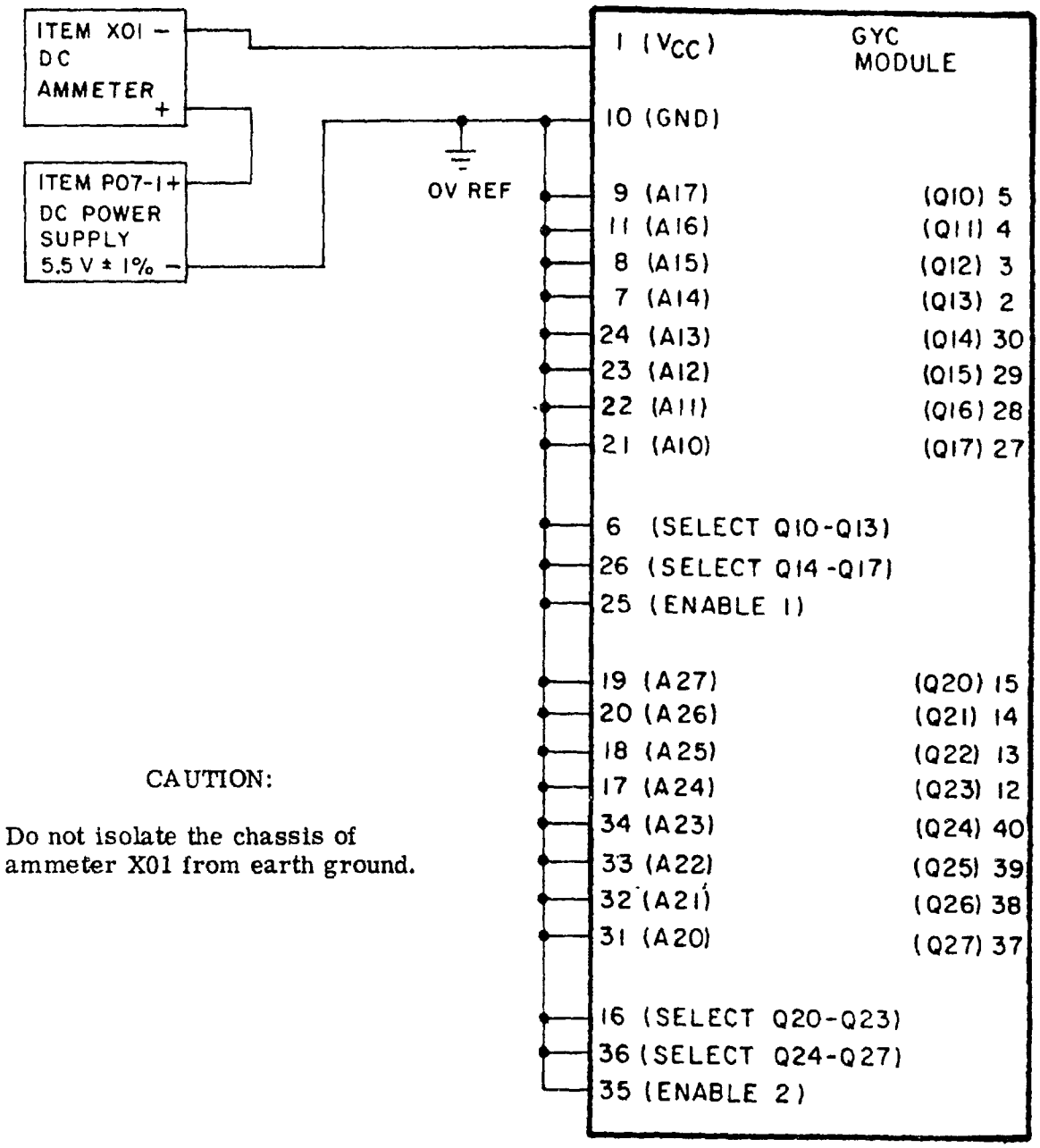
FIGURE 12. Input current (low-level and high-level) test circuit for table VII.

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TABLE VIII. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
H	Power supply current	10	450	mA	13

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 13. Power supply current test circuit for table VIII.

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TABLE IX. Output leakage current (high-impedance state) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Q10 Output leakage current (high-impedance state), V_O = 2.4 V	5	-100	100	µA	14
J2	Q11	4				
J3	Q12	3				
J4	Q13	2				
J5	Q14	30				
J6	Q15	29				
J7	Q16	28				
J8	Q17	27				
J9	Q24	40				
J10	Q25	39				
J11	Q26	38				
J12	Q27	37				
J13	Q20	15				
J14	Q21	14				
J15	Q22	13				
J16	Q23	12				

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TABLE IX. Output leakage current (high-impedance state) test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
J17	Q10 Output leakage current (high-impedance state), V_O = 0.45 V	5	-100	100	µA	14
J18	Q11	4				
J19	Q12	3				
J20	Q13	2				
J21	Q14	30				
J22	Q15	29				
J23	Q16	28				
J24	Q17	27				
J25	Q24	40				
J26	Q25	39				
J27	Q26	38				
J28	Q27	37				
J29	Q20	15				
J30	Q21	14				
J31	Q22	13				
J32	Q23	12				

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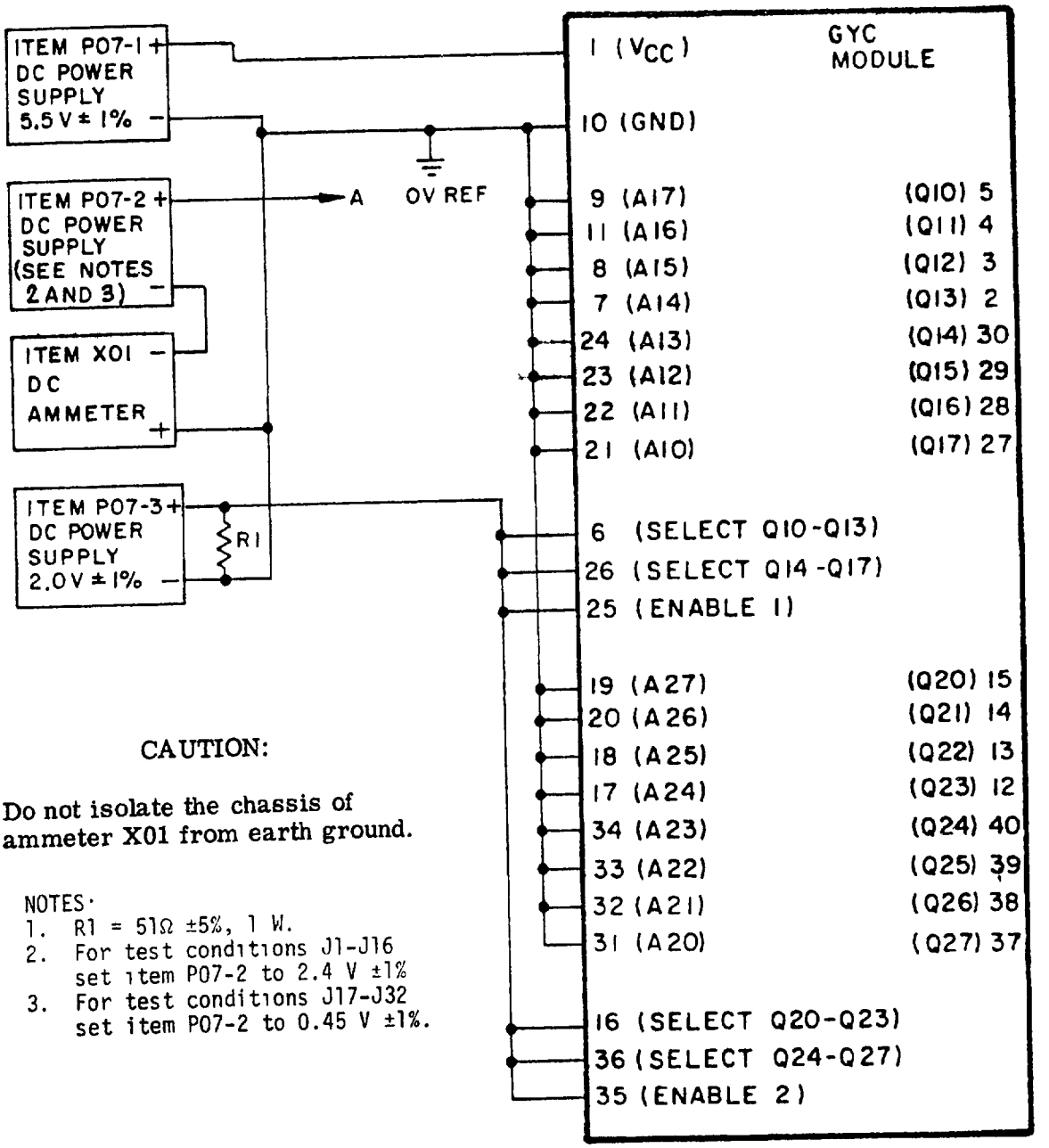


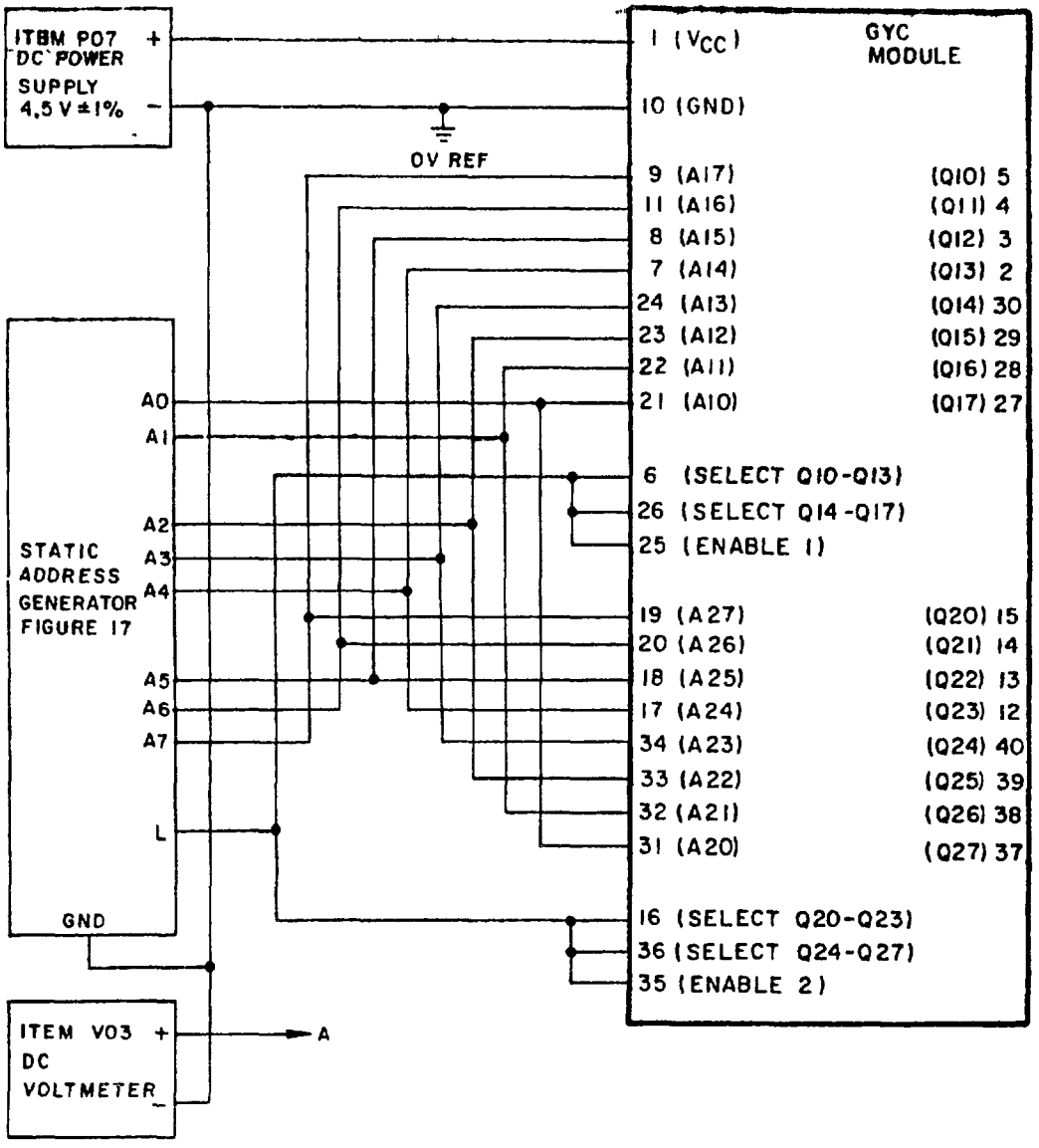
FIGURE 14. Output leakage current (high-impedance state) test circuit for table IX.

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TABLE X. Output voltage (Vo) level (unprogrammed) test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
K	Output voltage (V_O) level (unprogrammed)	← See note 1, figure 15 →			

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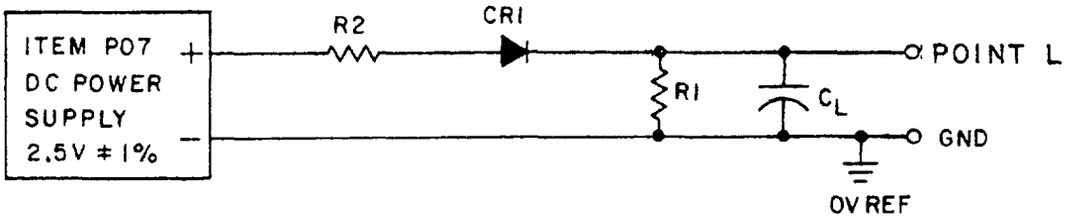


NOTES:

1. The static address generator is used to select, in turn, each of the 256 address combinations. For each address combination each output is monitored separately at point "A" to verify that all outputs have the voltage level as specified in the user's detailed module document for an unprogrammed module. If an output high level is specified, it shall be $\geq$ 2.4 V, and if a low level is specified, it shall be $\leq$ 0.45 V. An individual module shall have either a high level or a low level for all address combinations.
2. Load the output under test with a load as specified on figure 16.

FIGURE 15. Output voltage (V_O) level (unprogrammed) test circuit for table X

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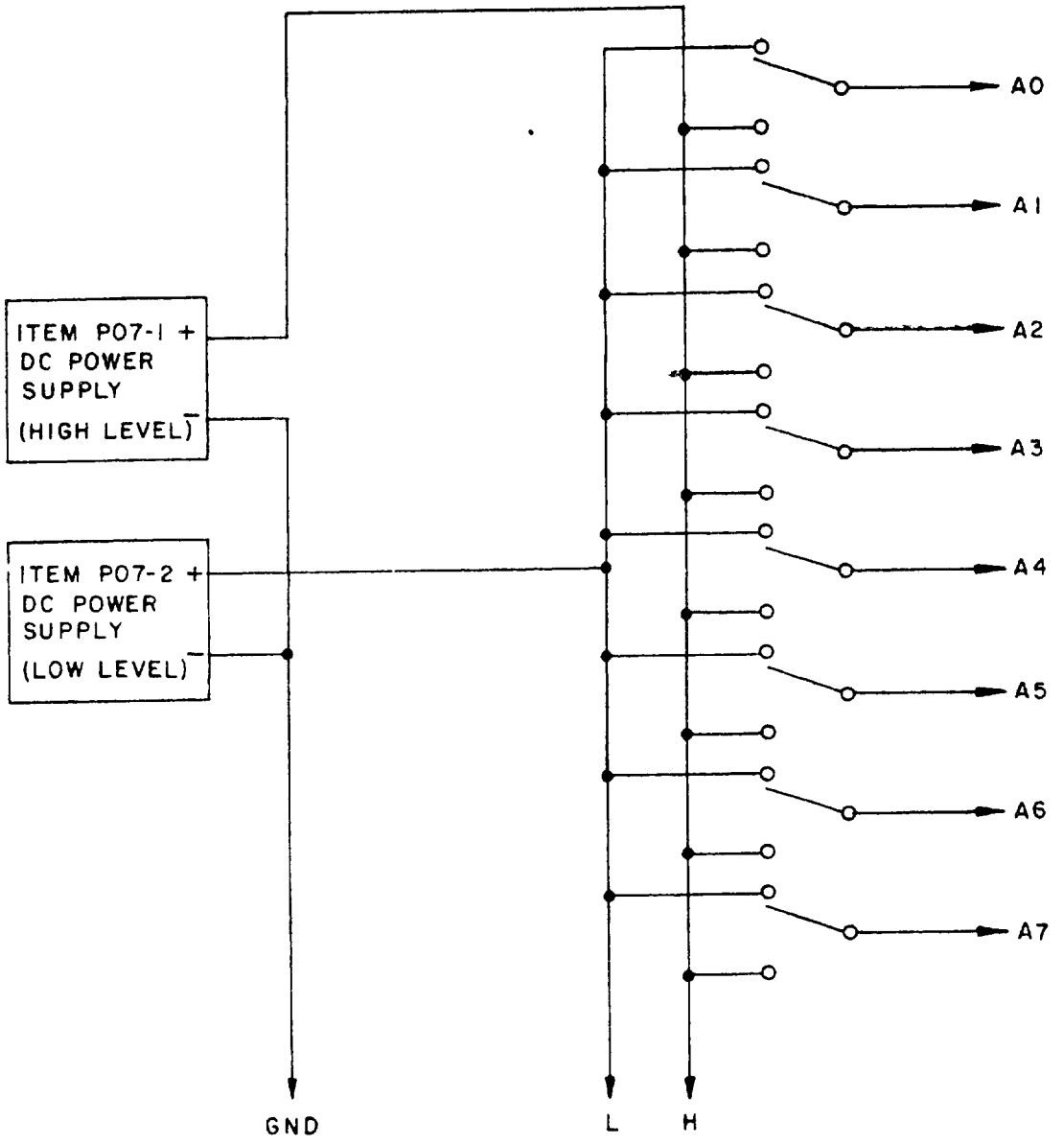


NOTES:

✳ 1. CR1 = 1N3600, 1N4150 or equivalent.
2. R1 = 2.67 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 10 mA ±5% out of point "L" with point "L" held at 0.45 V.
4. For test conditions R1-R16, S1-S16, V1, V2, W1-W6, and X1-X6, C_L = 50 ±5 pF including probe and test fixture.
5. For test conditions R17-R32, S17-S32, V3, and V4, C_L = 150 ±5 pF including probe and test fixture.
6. C_L is not required for dc electrical tests and test conditions T1-T32 and U1-U32.
7. During output switching the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 16. GYC logic load circuit.

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NOTES:
1. High level (H) is defined as 2.0 V ±1%.
2. Low level (L) is defined as 0.8 V ±1%.

FIGURE 17. Static address generator.

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TABLE XI. Power supply filter capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
L	Power supply filter capacitance	1	1.6	3.0	µF	18

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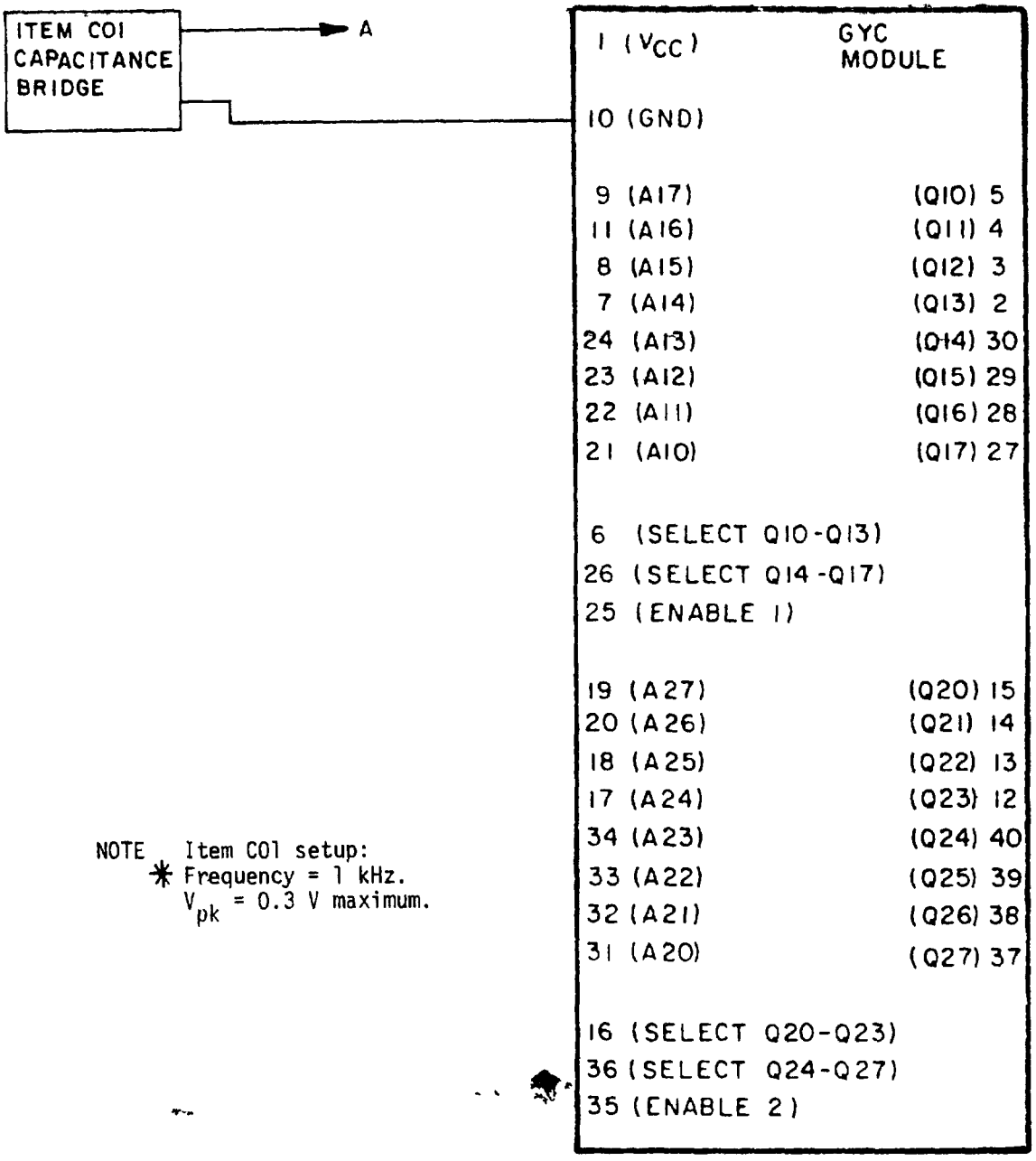


FIGURE 18. Power supply filter capacitance test circuit for table XI.

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TABLE XII. Programmability test.

Test condition	Parameter		Initial limits 25° ±5°C Ambient		Unit	Test figure
			Min	Max		
M	Programmability	The module shall be programmed in accordance with the user's detailed module document and the test pattern shall be as specified in table XIII. The step by step procedure as specified in the user's detailed module document shall be followed, including the program verification tests.	No failure			

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TABLE XIII. Memory qualification test pattern.

Address word	Memory outputs								Address word	Memory outputs							
	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7		Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7
000	L	L	L	L	L	L	L	L	025	H	H	L	L	H	H	L	L
001	H	H	H	H	H	H	H	H	026	L	H	L	H	L	H	L	H
002	L	L	L	H	L	L	L	H	027	L	L	H	H	L	L	H	H
003	H	H	H	H	H	H	H	H	028	H	L	L	H	H	L	L	H
004	L	L	H	L	L	L	H	L	029	L	H	H	L	L	H	H	L
005	L	H	L	H	L	H	L	H	030	H	L	H	L	H	L	H	L
006	L	H	H	L	L	H	H	L	031	L	L	L	L	L	L	L	L
007	L	L	L	L	L	L	L	L	032	L	H	L	H	L	H	L	H
008	L	L	H	H	L	L	H	H	033	H	H	L	L	H	H	L	L
009	H	L	L	H	H	L	L	H	034	L	H	H	L	L	H	H	L
010	H	L	H	L	H	L	H	L	035	L	L	L	L	L	L	L	L
011	H	H	L	L	H	H	L	L	036	H	H	H	H	H	H	H	H
012	L	L	H	H	L	L	H	H	037	H	H	H	H	H	H	H	H
013	L	H	H	L	L	H	H	L	038	H	H	H	H	H	H	H	H
014	H	L	H	L	H	L	H	L	039	H	H	H	H	H	H	H	H
015	L	L	L	L	L	L	L	L	040	L	L	H	H	L	L	H	H
016	L	H	L	L	L	H	L	L	041	H	L	L	H	H	L	L	H
017	L	H	L	H	L	H	L	H	042	L	L	H	H	L	L	H	H
018	H	L	L	H	H	L	L	H	043	L	L	L	L	L	L	L	L
019	H	H	L	L	H	H	L	L	044	H	H	L	L	H	H	L	L
020	L	H	L	H	L	H	L	H	045	H	L	H	L	H	L	H	L
021	H	L	H	L	H	L	H	L	046	L	H	H	L	L	H	H	L
022	L	L	H	H	L	L	H	H	047	L	L	H	H	L	L	H	H
023	L	L	L	L	L	L	L	L	048	H	L	H	L	H	L	H	L
024	L	H	H	L	L	H	H	L	049	L	H	L	H	L	H	L	H

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TABLE XIII. Memory qualification test pattern - Continued.

Address word	Memory outputs								Address word	Memory outputs							
	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7		Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7
050	H	H	L	L	H	H	L	L	075	H	H	H	H	H	H	H	H
051	L	L	L	L	L	L	L	L	076	H	L	H	L	H	L	H	L
052	H	L	L	H	H	L	L	H	077	L	L	L	L	L	L	L	L
053	H	L	H	L	H	L	H	L	078	H	H	L	L	H	H	L	L
054	L	L	H	H	L	I	H	H	079	L	L	H	H	L	L	H	H
055	L	H	L	H	L	H	L	H	080	L	H	H	L	L	H	H	L
056	H	L	L	H	H	L	L	H	081	L	H	L	H	L	H	L	H
057	L	H	H	L	L	H	H	L	082	H	L	H	L	H	L	H	L
058	H	H	L	L	H	H	L	L	083	H	L	L	H	H	L	L	H
059	L	L	L	L	L	L	L	L	084	H	H	L	L	H	H	L	L
060	L	L	H	H	L	L	H	H	085	L	L	L	L	L	L	L	L
061	H	L	L	H	H	L	L	H	086	H	L	L	H	H	L	L	H
062	L	H	L	H	L	H	L	H	087	L	H	H	L	L	H	H	L
063	H	H	L	L	H	H	L	L	088	L	L	H	H	L	L	H	H
064	L	H	H	L	L	H	H	L	089	L	H	L	H	L	H	L	H
065	H	L	L	H	H	L	L	H	090	H	L	H	L	H	L	H	L
066	H	L	H	L	H	L	H	L	091	H	H	L	L	H	H	L	L
067	L	L	H	H	L	L	H	H	092	L	H	H	L	L	H	H	L
068	L	H	L	H	L	H	L	H	093	L	L	L	L	L	L	L	L
069	L	L	L	L	L	L	L	L	094	L	H	L	H	L	H	L	H
070	L	H	H	L	L	H	H	L	095	L	L	H	H	L	L	H	H
071	H	L	L	H	H	L	L	H	096	H	L	L	H	H	L	L	H
072	H	H	H	H	H	H	H	H	097	L	L	L	L	L	L	L	L
073	H	H	H	H	H	H	H	H	098	H	H	L	L	H	H	L	L
074	H	H	H	H	H	H	H	H	099	H	L	H	L	H	L	H	L

See footnote at end of table.

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TABLE XIII. Memory qualification test pattern - Continued.

Address word	Memory outputs								Address word	Memory outputs							
	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7		Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7
100	H	H	L	L	H	H	L	L	125	H	L	H	L	H	L	H	L
101	H	L	L	H	H	L	L	H	126	L	H	L	H	L	H	L	H
102	L	H	L	H	L	H	L	H	127	L	L	H	H	L	L	H	H
103	L	L	H	H	L	L	H	H	128	L	H	H	H	L	H	H	H
104	H	L	H	L	H	L	H	L	129	H	L	L	H	H	L	L	H
105	L	L	L	L	L	L	L	L	130	L	H	H	L	L	H	H	L
106	H	H	L	L	H	H	L	L	131	H	L	H	L	H	L	H	L
107	L	H	L	H	L	H	L	H	132	H	H	L	L	H	H	L	L
108	H	H	H	H	H	H	H	H	133	L	H	H	L	L	H	H	L
109	H	H	H	H	H	H	H	H	134	L	L	L	L	L	L	L	L
110	H	H	H	H	H	H	H	H	135	L	L	H	H	L	L	H	H
111	H	H	H	H	H	H	H	H	136	L	H	L	H	L	H	L	H
112	L	L	H	H	L	L	H	H	137	H	L	H	L	H	L	H	L
113	L	L	L	L	L	L	L	L	138	H	H	L	L	H	H	L	L
114	L	H	H	L	L	H	H	L	139	H	L	L	H	H	L	L	H
115	H	L	H	L	H	L	H	L	140	L	H	L	H	L	H	L	H
116	H	H	L	L	H	H	L	L	141	L	L	H	H	L	L	H	H
117	L	H	L	H	L	H	L	H	142	L	L	L	L	L	L	L	L
118	L	L	H	H	L	L	H	H	143	H	L	H	L	H	L	H	L
119	H	L	L	H	H	L	L	H	144	H	H	H	H	H	H	H	H
120	H	L	H	L	H	L	H	L	145	H	H	H	H	H	H	H	H
121	L	L	L	L	L	L	L	L	146	H	H	H	H	H	H	H	H
122	L	H	H	L	L	H	H	L	147	H	H	H	H	H	H	H	H
123	L	L	H	H	L	L	H	H	148	L	H	H	L	L	H	H	L
124	H	H	L	L	H	H	L	L	149	H	H	L	L	H	H	L	L

See footnote at end of table.

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TABLE XIII. Memory qualification test pattern - Continued.

Address word	Memory outputs								Address word	Memory outputs							
	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7		Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7
150	L	L	L	L	L	L	L	L	175	L	H	L	H	L	H	L	H
151	H	L	L	H	H	L	L	H	176	H	L	H	L	H	L	H	L
152	L	L	H	H	L	L	H	H	177	H	H	L	L	H	H	L	L
153	H	L	L	H	H	L	L	H	178	L	L	L	L	L	L	L	L
154	L	H	L	H	L	H	L	H	179	L	H	H	L	L	H	H	L
155	L	H	H	L	L	H	H	L	180	H	H	H	H	H	H	H	H
156	H	L	H	L	H	L	H	L	181	H	H	H	H	H	H	H	H
157	H	H	L	L	H	H	L	L	182	H	H	H	H	H	H	H	H
158	L	L	L	L	L	L	L	L	183	H	H	H	H	H	H	H	H
159	H	L	L	H	H	L	L	H	184	L	H	L	H	L	H	L	H
160	L	I	H	H	L	L	H	H	185	L	L	H	H	L	L	H	H
161	L	H	L	H	L	H	L	H	186	L	L	L	L	L	L	L	L
162	L	L	L	L	L	L	L	L	187	H	L	L	H	H	L	L	H
163	L	H	H	L	L	H	H	L	188	L	H	L	H	L	H	L	H
164	H	L	H	L	H	L	H	L	189	L	H	H	L	L	H	H	L
165	H	H	L	L	H	H	L	L	190	H	L	H	L	H	L	H	L
166	H	L	H	L	H	L	H	L	191	H	H	L	L	H	H	L	L
167	L	L	H	H	L	L	H	H	192	H	H	H	H	H	H	H	H
168	L	H	H	L	L	H	H	L	193	H	H	H	H	H	H	H	H
169	L	H	L	H	L	H	L	H	194	H	H	H	H	H	H	H	H
170	L	L	L	L	L	L	L	L	195	L	L	H	H	L	L	H	H
171	H	H	L	L	H	H	L	L	196	L	L	L	L	L	L	L	L
172	H	L	H	L	H	L	H	L	197	L	H	L	H	L	H	L	H
173	L	L	H	H	L	L	H	H	198	H	L	H	L	H	L	H	L
174	L	H	H	L	L	H	H	L	199	L	H	H	L	L	H	H	L

See footnote at end of table.

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TABLE XIII. Memory qualification test pattern - Continued.

Address word	Memory outputs								Address word	Memory outputs							
	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7		Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7
200	H	L	L	H	H	L	L	H	225	L	H	H	L	L	H	H	L
201	H	H	L	L	H	H	L	L	226	H	H	L	L	H	H	L	L
202	H	L	H	L	H	L	H	L	227	H	L	L	H	H	L	L	H
203	L	L	H	H	L	L	H	H	228	H	L	H	L	H	L	H	L
204	L	L	L	L	L	L	L	L	229	L	L	H	H	L	L	H	H
205	L	H	L	H	L	H	L	H	230	L	H	H	L	L	H	H	L
206	H	L	L	H	H	L	L	H	231	L	H	L	H	L	H	L	H
207	L	H	H	L	L	H	H	L	232	L	L	L	L	L	L	L	L
208	H	H	L	L	H	H	L	L	233	H	H	L	L	H	H	L	L
209	H	L	H	L	H	L	H	L	234	H	L	H	L	H	L	H	L
210	L	L	H	H	L	L	H	H	235	H	L	L	H	H	L	L	H
211	L	H	L	H	L	H	L	H	236	L	H	H	L	L	H	H	L
212	L	L	L	L	L	L	L	L	237	L	H	L	H	L	H	L	H
213	H	L	H	L	H	L	H	L	238	L	L	H	H	L	L	H	H
214	L	H	H	L	L	H	H	L	239	H	L	H	L	H	L	H	L
215	H	L	L	H	H	L	L	H	240	L	L	L	L	L	L	L	L
216	H	H	H	H	H	H	H	H	241	L	H	H	L	L	H	H	L
217	H	H	H	H	H	H	H	H	242	L	H	L	H	L	H	L	H
218	H	H	H	H	H	H	H	H	243	H	L	L	H	H	L	L	H
219	H	H	H	H	H	H	H	H	244	H	L	H	L	H	L	H	L
220	L	L	L	L	L	L	L	L	245	L	L	H	H	L	L	H	H
221	H	H	L	L	H	H	L	L	246	L	H	L	H	L	H	L	H
222	L	L	H	H	L	L	H	H	247	H	L	L	H	H	L	L	H
223	L	H	L	H	L	H	L	H	248	L	L	L	L	L	L	L	L
224	L	L	L	L	L	L	L	L	249	L	H	H	L	L	H	H	L

See footnote at end of table.

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TABLE XIII. Memory qualification test pattern - Continued.

Address word	Memory outputs								Address word	Memory outputs							
	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7		Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7
250	H	H	L	L	H	H	L	L	253	H	H	H	H	H	H	H	H
251	L	L	H	H	L	L	H	H	254	H	H	H	H	H	H	H	H
252	H	H	H	H	H	H	H	H	255	H	H	H	H	H	H	H	H

1/ Program each of the two independent 256-word by 8-bit PROM with this test pattern.

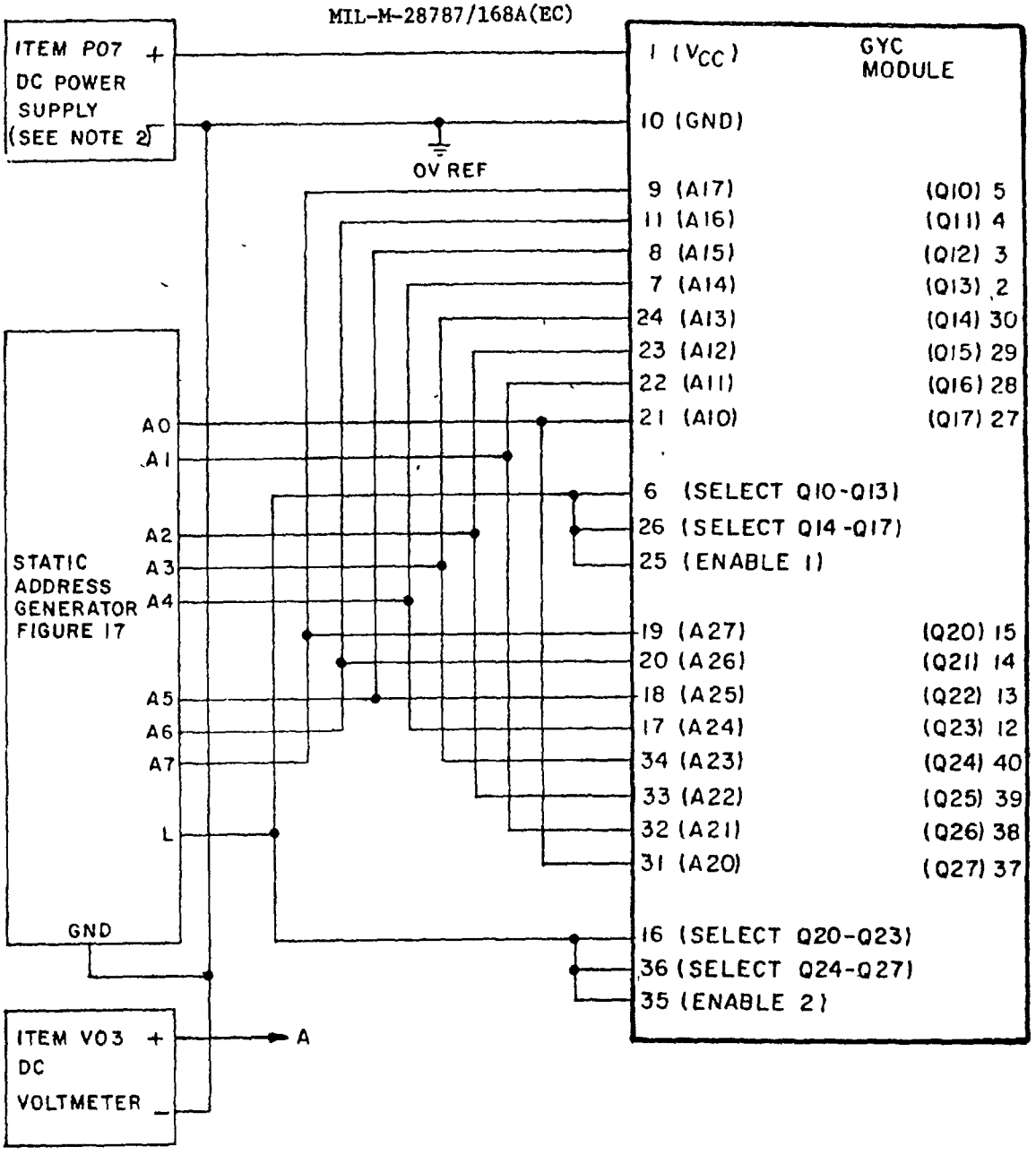
L = Low-level output.
H = High-level output.

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TABLE XIV. Functional test pattern test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
N	Functional test pattern	← See note 1, figure 19 →			

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NOTES:

1. The static address generator is used to select, in turn, each of the 256 address combinations. For each address combination the output is monitored at point "A" to verify the test pattern of table XIII. Output high level shall be $\geq$ 2.4 V and output low level shall be $\leq$ 0.45 V.
2. Adjust item P07 either as specified in the user's detailed module document or, if not specified, to 4.5 V ±1%.
3. Load the output under test either with the load as specified in the user's detailed module document or, if not specified, with the load as specified on figure 16.
4. The purpose of notes 2 and 3 is to allow the user to select a V_{CC} voltage and a load current which will insure the PROM fuse link has been blown.

FIGURE 19. Functional test pattern test circuit for table XIV.

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TABLE XV. Static output voltage (low-level and high-level) test.

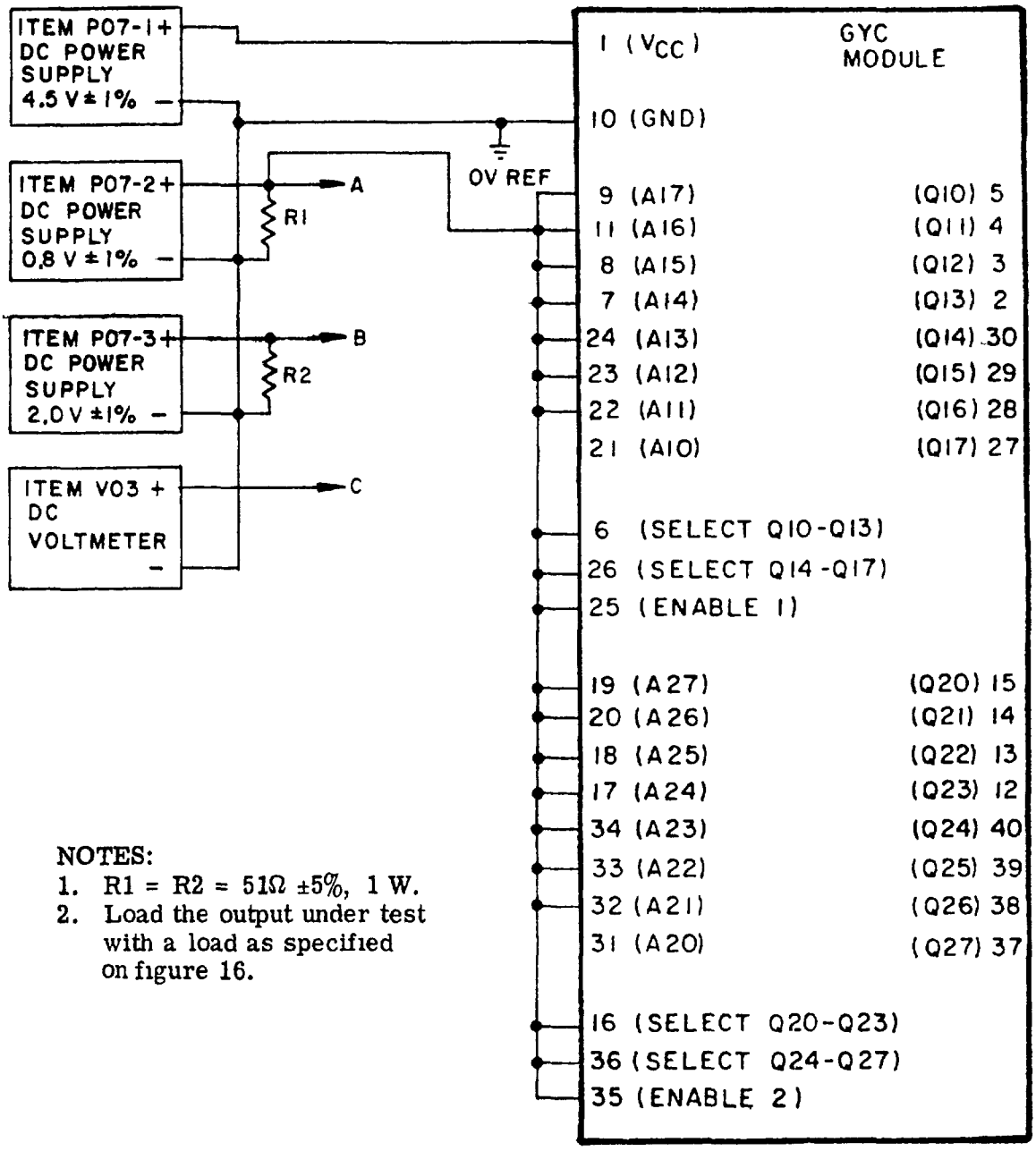
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	Q10	Static output voltage, low level	21	Open	5	0	0.45	V	20
P2	Q11				4				
P3	Q12				3				
P4	Q13				2				
P5	Q14				30				
P6	Q15				29				
P7	Q16				28				
P8	Q17				27				
P9	Q24		31		40				
P10	Q25				39				
P11	Q26				38				
P12	Q27				37				
P13	Q20				15				
P14	Q21				14				
P15	Q22				13				
P16	Q23				12				

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TABLE XV. Static output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q1	Q10	Static output voltage, high level	Open	21	5	2.4	4.5	V	20
Q2	Q11				4				
Q3	Q12				3				
Q4	Q13				2				
Q5	Q14				30				
Q6	Q15				29				
Q7	Q16				28				
Q8	Q17				27				
Q9	Q24			31	40				
Q10	Q25				39				
Q11	Q26				38				
Q12	Q27				37				
Q13	Q20				15				
Q14	Q21				14				
Q15	Q22				13				
Q16	Q23				12				

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load the output under test with a load as specified on figure 16.

FIGURE 20. Static output voltage (low-level and high-level) test circuit for table XV.

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
R1	Q10	Transition time (t_{TLH}), C_L = 50 pF	5	5	30	ns	21,22
R2	Q11		4				
R3	Q12		3				
R4	Q13		2				
R5	Q14		30				
R6	Q15		29				
R7	Q16		28				
R8	Q17		27				
R9	Q24		40				
R10	Q25		39				
R11	Q26		38				
R12	Q27		37				
R13	Q20		15				
R14	Q21		14				
R15	Q22		13				
R16	Q23		12				
R17	Q10	Transition time (t_{TLH}), C_L = 150 pF	5		40		
R18	Q11		4				
R19	Q12		3				
R20	Q13		2				
R21	Q14		30				
R22	Q15		29				
R23	Q16		28				
R24	Q17		27				

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test - Continued.

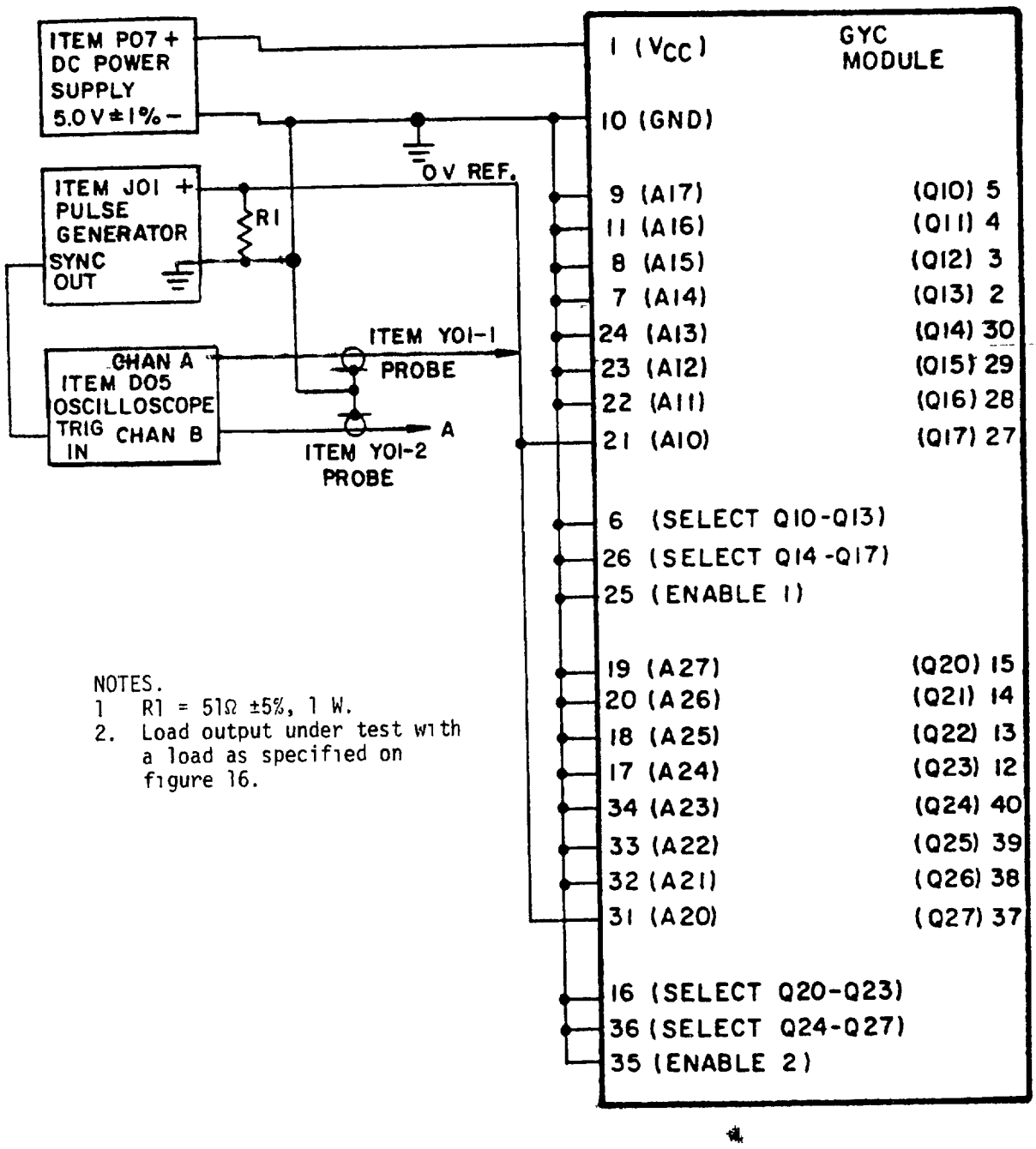
Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
R25	Q24 Transition time (t_{TLH}), C_L = 150 pF	40	5	40	ns	21,22
R26	Q25	39				
R27	Q26	38				
R28	Q27	37				
R29	Q20	15				
R30	Q21	14				
R31	Q22	13				
R32	Q23	12				
S1	Q10 Transition time (t_{THL}), C_L = 50 pF	5	3	30		
S2	Q11	4				
S3	Q12	3				
S4	Q13	2				
S5	Q14	30				
S6	Q15	29				
S7	Q16	28				
S8	Q17	27				
S9	Q24	40				
S10	Q25	39				
S11	Q26	38				
S12	Q27	37				
S13	Q20	15				
S14	Q21	14				
S15	Q22	13				
S16	Q23	12				

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
S17	Q10 Transition time (t_{THL}), C_L = 150 pF	5	5	40	ns	21,22
S18	Q11	4	↓	↓	↓	↓
S19	Q12	3				
S20	Q13	2				
S21	Q14	30				
S22	Q15	29				
S23	Q16	28				
S24	Q17	27				
S25	Q24	40				
S26	Q25	39				
S27	Q26	38				
S28	Q27	37				
S29	Q20	15				
S30	Q21	14				
S31	Q22	13				
S32	Q23	12				

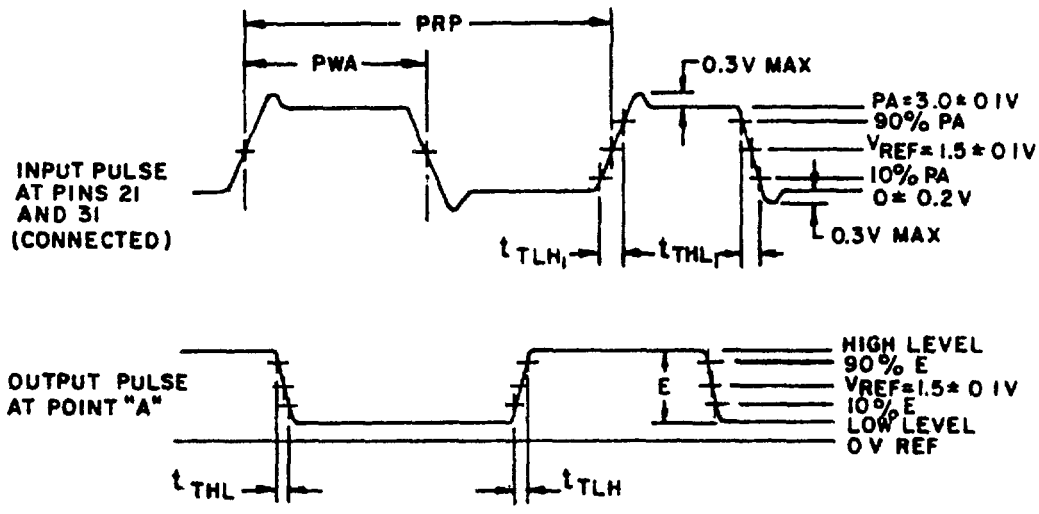
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NOTES.
1. R1 = 51Ω ±5%, 1 W.
2. Load output under test with a load as specified on figure 16.

* FIGURE 21. Transition time (t_{TLH} and t_{THL}) test circuit for table XVI.

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NOTES
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 200 ±50 ns.
   PRP = 400 ±50 ns.
2. Before making the measurements specified in table XVI, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.45 V.
3. All voltages are referenced to ground.

FIGURE 22. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XVI.

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TABLE XVII. Enable/select recovery time test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	Q10 Enable recovery time (t_{ERL}), E1, C_L <10 pF	16,36,35	25	5	6,26,21,31	5	40	ns	23,24
T2	Q11			4					
T3	Q12			3					
T4	Q13			2					
T5	Q14			30					
T6	Q15			29					
T7	Q16			28					
T8	Q17			27					
T9	Q24 Enable recovery time (t_{ERL}), E2, C_L <10 pF	25,26,6	35	40	16,36,21,31				
T10	Q25			39					
T11	Q26			38					
T12	Q27			37					
T13	Q20			15					
T14	Q21			14					
T15	Q22			13					
T16	Q23			12					
T17	Q10 Enable recovery time (t_{ERL}), select Q10-Q13	16,36,35	6	5	26,25,21,31				
T18	Q11 C_L <10 pF			4					
T19	Q12			3					
T20	Q13			2					

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TABLE XVII Enable/select recovery time test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T21	Q14 Enable recovery time (t_{ERL}), select Q14-Q17,	16,36,35	26	30	6,25,21,31	5	40	ns	23,24
T22	Q15 C_L <10pF			29					
T23	Q16			28					
T24	Q17			27					
T25	Q24 Enable recovery time (t_{ERL}), select Q24-Q27,	6,26,25	36	40	16,35,21,31				
T26	Q25 C_L <10pF			39					
T27	Q26			38					
T28	Q27			37					
T29	Q20 Enable recovery time (t_{ERL}), select Q20-Q23,		16	15	36,35,21,31				
T30	Q21 C_L <10pF			14					
T31	Q22			13					
T32	Q23			12					
U1	Q10 Enable recovery time (t_{ERH}), E1, C_L <10pF	16,36,35, 21,31	25	5	6,26				
U2	Q11			4					
U3	Q12			3					
U4	Q13			2					
U5	Q14			30					
U6	Q15			29					
U7	Q16			28					
U8	Q17			27					

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TABLE XVII. Enable/select recovery time test - Continued.

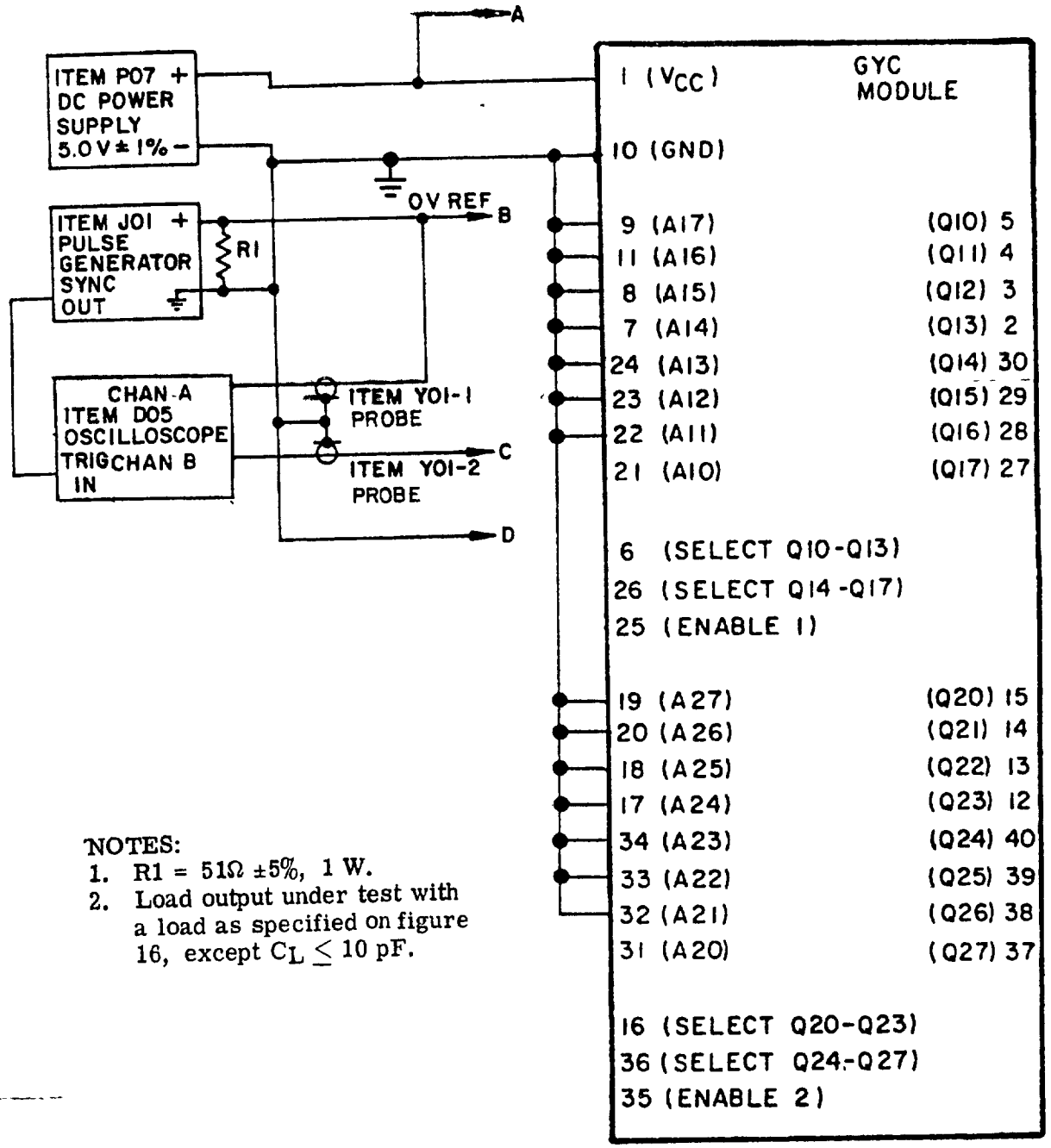
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U9	Q24 Enable recovery time (t_{ERH}), E2, C_L <10pF	6,26,25,21, 31	35	40	16,36	5	40	ns	23,24
U10	Q25			39					
U11	Q26			38					
U12	Q27			37					
U13	Q20			15					
U14	Q21			14					
U15	Q22			13					
U16	Q23			12					
U17	Q10 Enable recovery time (t_{ERH}), select Q10-Q13,	16,36,35, 21,31	6	5	26,25				
U18	Q11 C_L <10pF			4					
U19	Q12			3					
U20	Q13			2					
U21	Q14 Enable recovery time (t_{ERH}), select Q14-Q17,		26	30	6,25				
U22	Q15 C_L <10pF			29					
U23	Q16			28					
U24	Q17			27					
U25	Q24 Enable recovery time (t_{ERH}), select Q24-Q27,	6,26,25,21, 31	36	40	16,35				
U26	Q25 C_L <10pF			39					
U27	Q26			38					

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TABLE XVII. Enable/select recovery time test - Continued

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U28	Q27	Enable recovery time (t_{ERH}), select Q24-Q27, C_L <10 pF	6,26,25,21, 31	36	37	16,35	5	40	ns	23,24
U29	Q20	Enable recovery time (t_{ERH}), select Q20-Q23, C_L <10 pF		16	15	36,35				
U30	Q21				14					
U31	Q22				13					
U32	Q23				12					

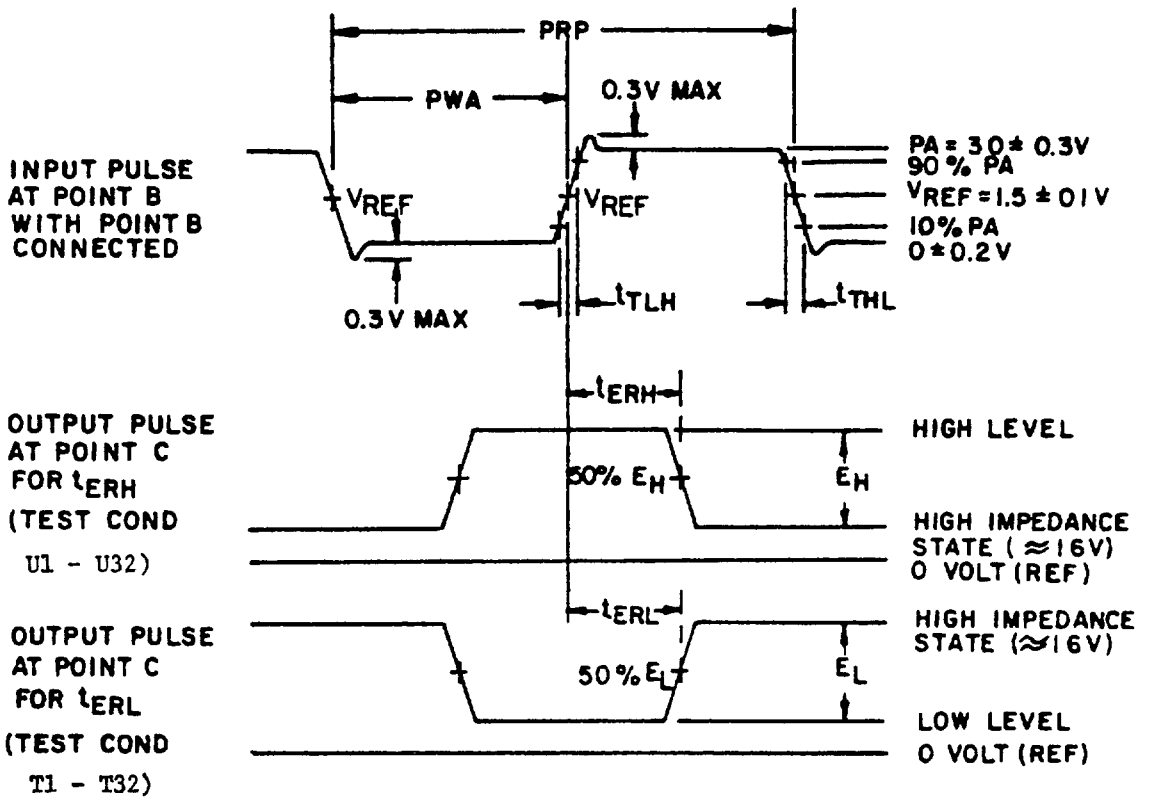
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load output under test with a load as specified on figure 16, except $C_L \leq 10$ pF.

✱ FIGURE 23. Enable/select recovery time test circuit for table XVII.

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NOTES:
1. Input pulse parameters
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the measurements specified in table XVII, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.45 V.
3. The lead length of electrical conductors between resistor R1 and input pin and between probes Y01-1 and Y01-2 and input and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

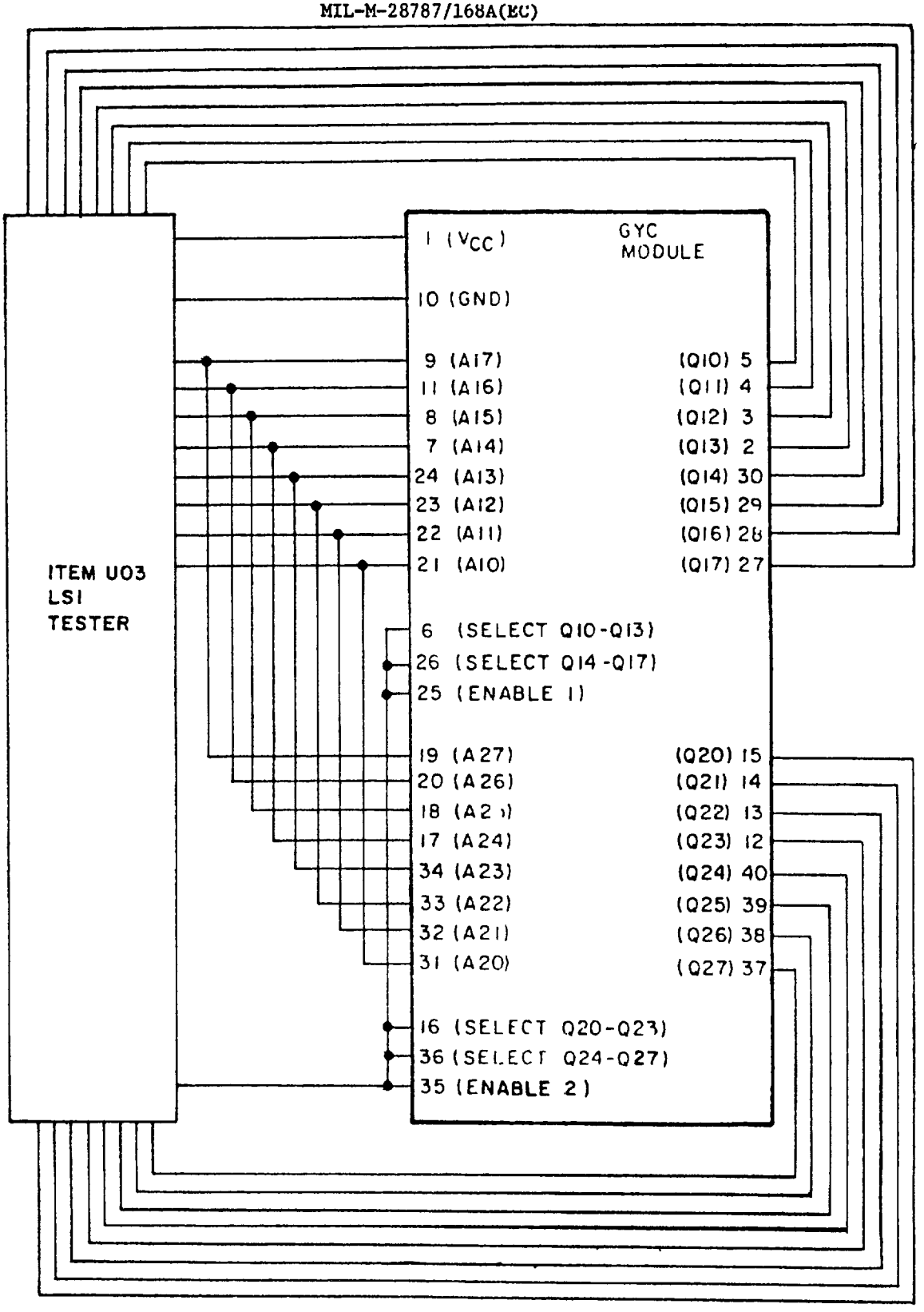
FIGURE 24. Enable/select recovery time test circuit waveforms for table XVII.

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TABLE XVIII. Galloping read access pattern test.

Test condition	Parameter		Initial limits 25° ±5°C Ambient		Unit	Test figure
			Min	Max		
V1	Galloping read access pattern (C_L = 50 pF, V_{CC} = 4.5 V)	Module shall conform to the requirements of figures 25 and 26 for each test condition.	No failure			
V2	Galloping read access pattern (C_L = 50 pF, V_{CC} = 5.5 V)		No failure			
V3	Galloping read access pattern (C_L = 150 pF, V_{CC} = 4.5 V)		No failure			
V4	Galloping read access pattern (C_L = 150 pF, V_{CC} = 5.5 V)		No failure			

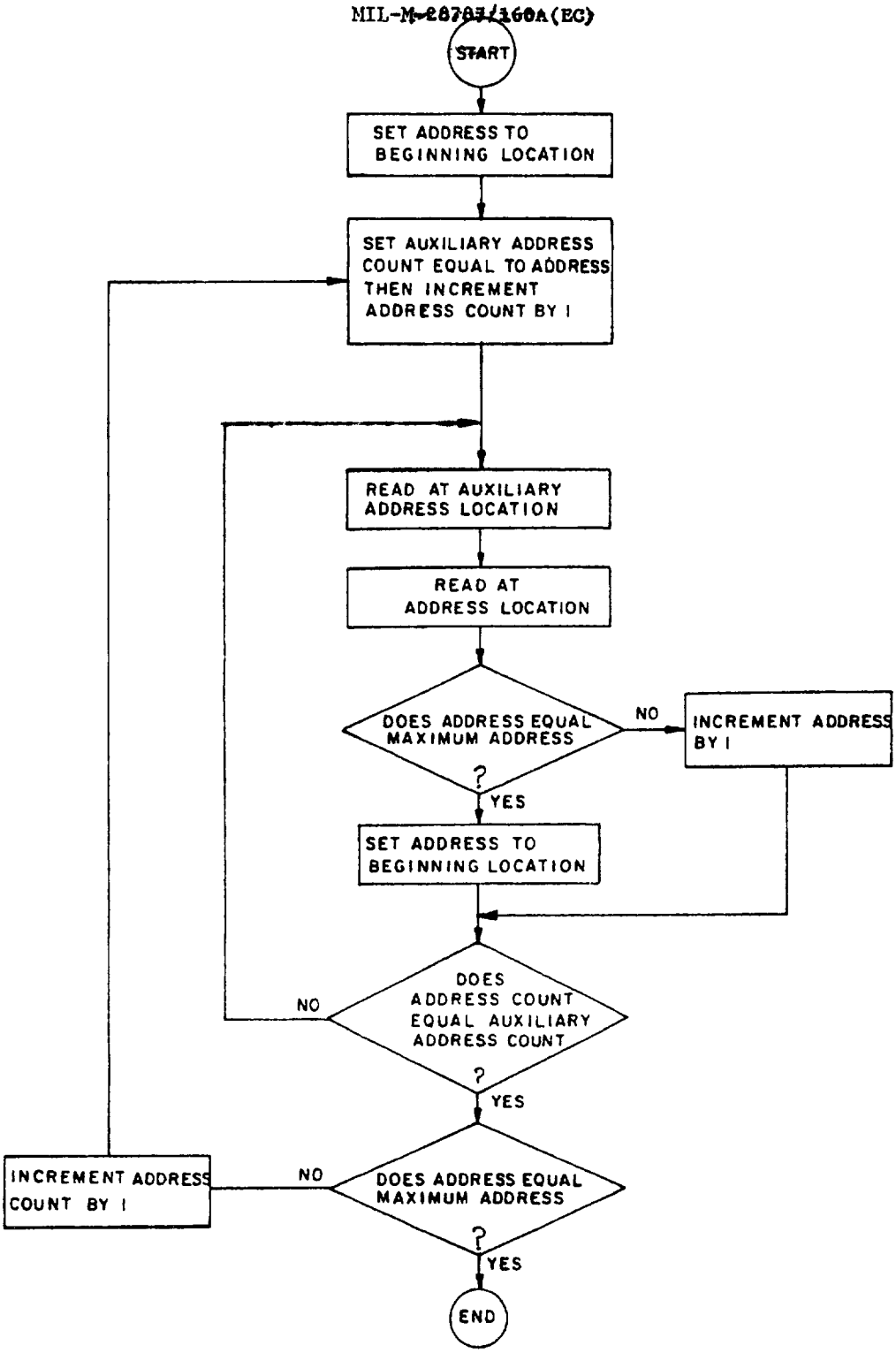
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NOTES
1. V_{CC} = 4.5 V ±1% for test conditions V1 and V3
2 V_{CC} = 5.5 V ±1% for test conditions V2 and V4.
3. Load output under test with a load as specified on figure 16.

FIGURE 25. Galloping read access pattern test circuit for table XVIII.

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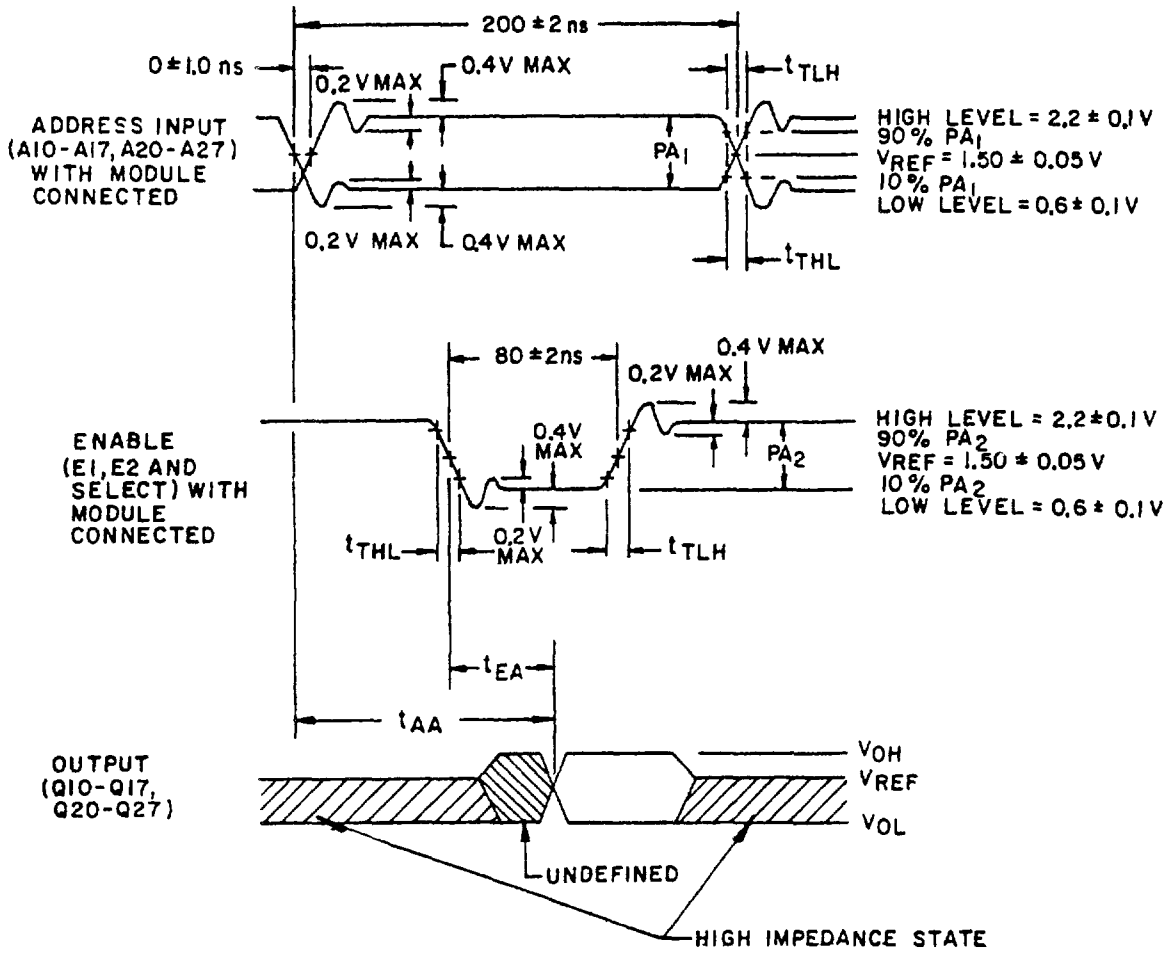


GALLOPING READ ACCESS PATTERN PROGRAM FLOW CHART

NOTE: Pattern is read and verified at each location.

FIGURE 26. Galloping read access pattern program flow chart, waveforms, and program for table XVIII.

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READ ACCESS INPUT AND OUTPUT WAVEFORMS

NOTES
1. High-impedance state = 1.6 V.
2. Address access time (50 pF load) $t_{AA1} = 80 \, ^{+0}_{-4}$ ns.
3. Address access time (150 pF load) $t_{AA2} = 90 \, ^{+0}_{-4}$ ns.
4. Enable access time (50 pF load) $t_{EA1} = 45 \, ^{+0}_{-2}$ ns.
5. Enable access time (150 pF load) $t_{EA2} = 55 \, ^{+0}_{-2}$ ns.
6. $t_{TLH} = t_{THL} = 10 \pm 4$ ns.

FIGURE 26. Galloping read access pattern program flow chart, waveforms, and program for table XVIII - Continued.

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Program for test conditions V1-V4:

1. The objective of the galloping read access pattern is to test address access time. It is necessary to test all possible address transitions. The test also verifies circuit enable time (t_{EA}) and address access time (t_{AA}).

2. A program that fulfills the galloping read access test objective would be:

(a) Read memory address zero.
(b) Read memory address one.
(c) Read memory address zero.
(d) Read memory address two.
(e) Read memory address zero.
(f) Read memory address three.
.
.
.

(g) Read memory address zero.
(h) Read memory address 255.
(i) Read memory address one.
(j) Read memory address two.
(k) Read memory address one.
(l) Read memory address three.
.
.
.

(m) Read memory address one.
(n) Read memory address 255.
(o) Read memory address one.
(p) Read memory address zero.
(q) Read memory address two.
(r) Read memory address three.
(s) Read memory address two.
(t) Read memory address four.
.
.
.

(u) Continue this pattern until all memory addresses have been tested to all other memory addresses.

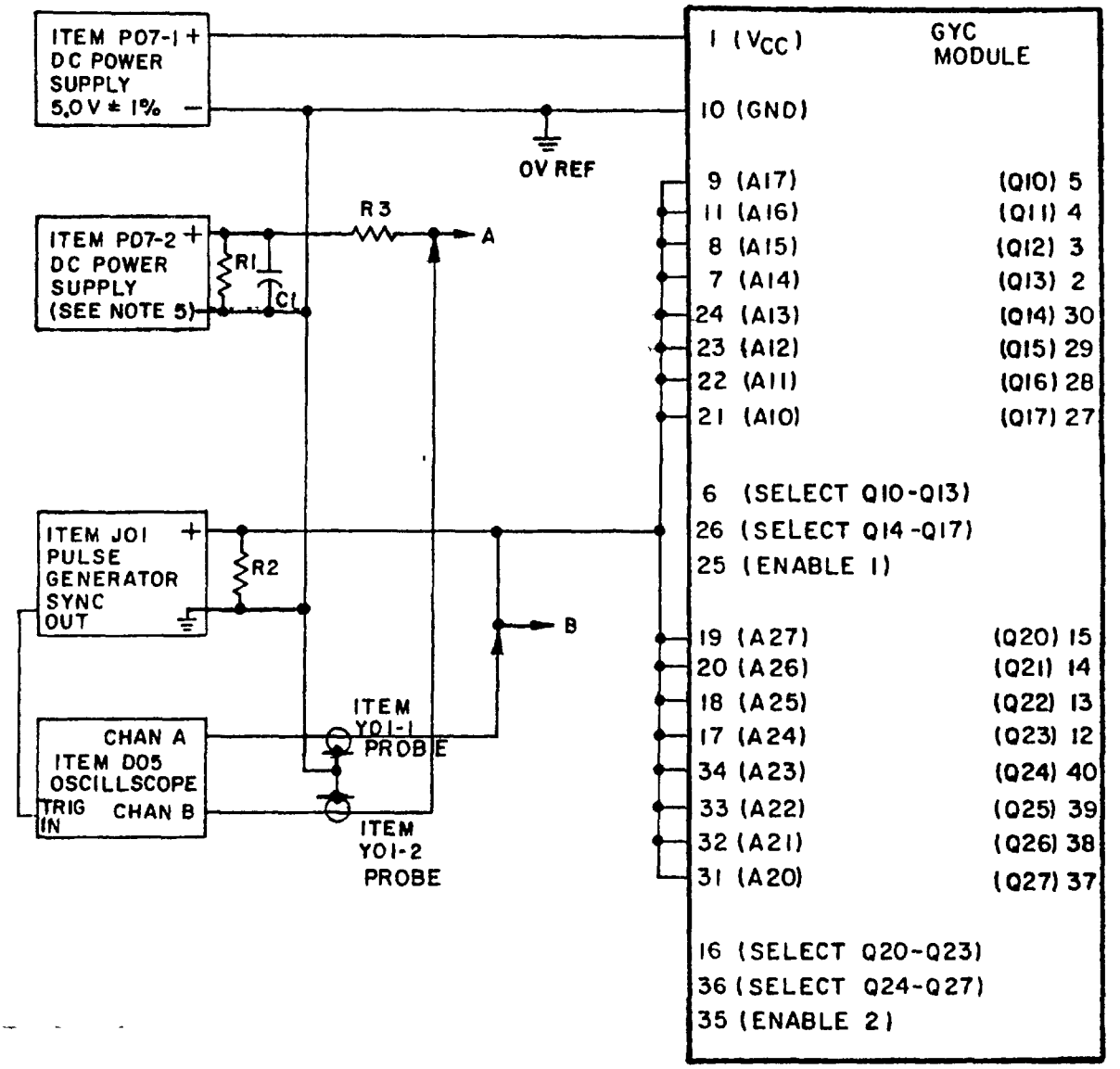
FIGURE 26. Galloping read access pattern program flow chart, waveforms, and program for table XVIII - Continued.

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TABLE XIX. Input crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W1	Enable 1	Input crosstalk, low level	25	35,6,26,16,36	0	400	mV	27,28
W2	Enable 2	↓	35	25,6,26,16,36	↓	↓	↓	↓
W3	Select Q10-Q13	↓	6	25,35,26,16,36	↓	↓	↓	↓
W4	Select Q14-Q17	↓	26	25,35,6,16,36	↓	↓	↓	↓
W5	Select Q20-Q23	↓	16	25,35,6,26,36	↓	↓	↓	↓
W6	Select Q24-Q27	↓	36	25,35,6,26,16	↓	↓	↓	↓
X1	Enable 1	Input crosstalk, high level	25	35,6,26,16,36	↓	↓	↓	↓
X2	Enable 2	↓	35	25,6,26,16,36	↓	↓	↓	↓
X3	Select Q10-Q13	↓	6	25,35,26,16,36	↓	↓	↓	↓
X4	Select Q14-Q17	↓	26	25,35,6,16,36	↓	↓	↓	↓
X5	Select Q20-Q23	↓	16	25,35,6,26,36	↓	↓	↓	↓
X6	Select Q24-Q27	↓	36	25,35,6,26,16	↓	↓	↓	↓

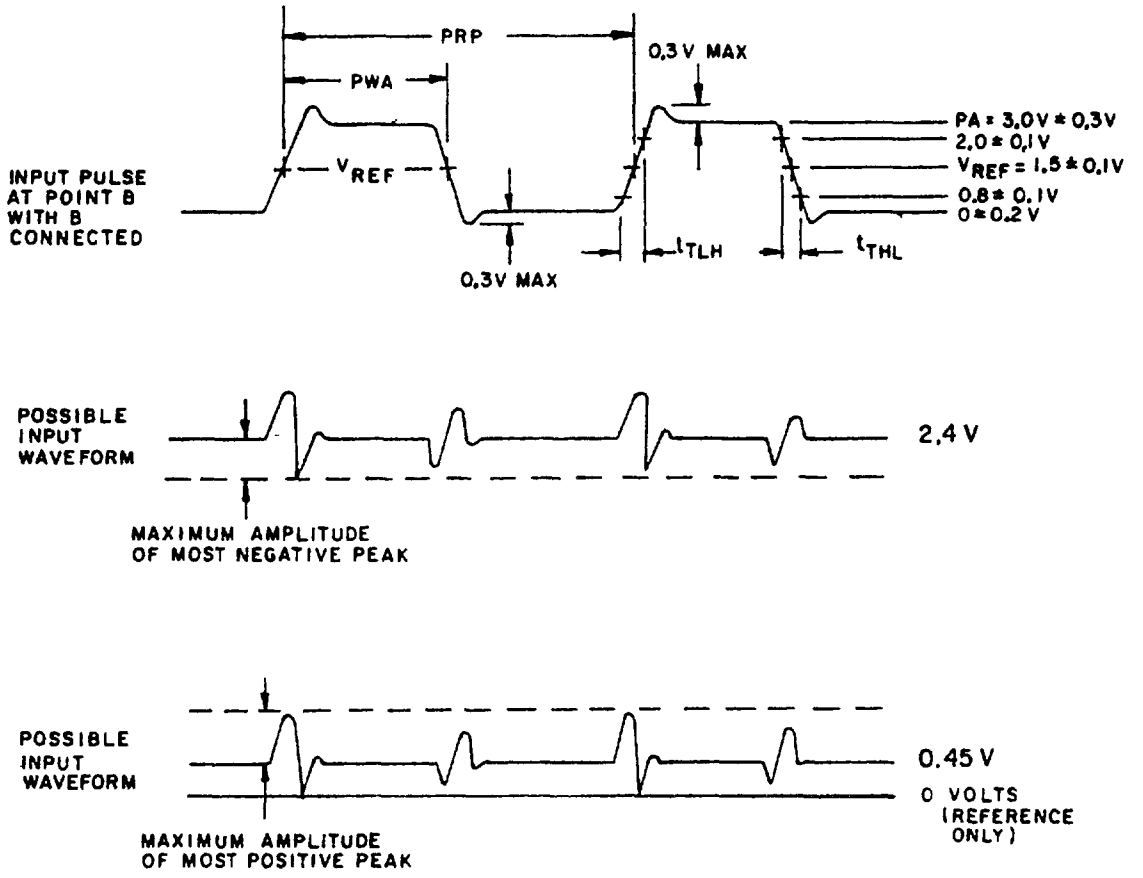
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 150Ω ±1%, 1/8 W.
3. C1 = Add sufficient capacitance to reduce item P07-2 power supply noise to <50 mV.
4. Load all outputs with a load as specified on figure 16.
5. For test conditions W1-W6, set item P07-2 to 0.45 V ±1%. For test conditions X1-X6, set item P07-2 to 2.4 V ±1%.

FIGURE 27. Input crosstalk (low-level and high-level) test circuit for table XIX.

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NOTES:

1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns verified at each input.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors shall be less than 2 inches (50.8 mm) between: resistors and input pins; probes Y01-1 and Y01-2 and input pins; load circuits and output pins; ground leads of probes Y01-1 and Y01-2 and test fixture and ground pins.

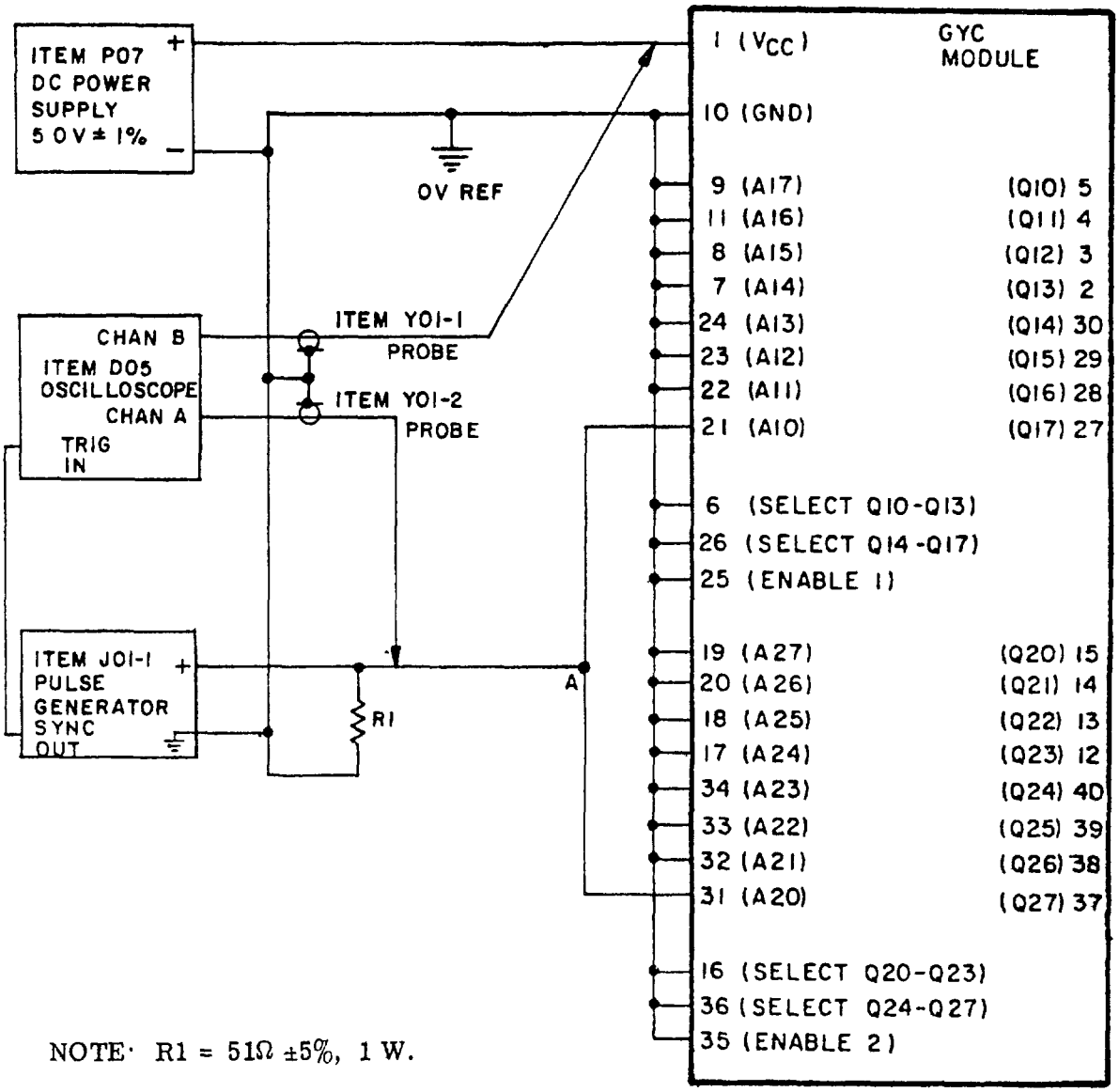
FIGURE 28. Input crosstalk (low-level and high-level) test circuit waveforms for table XIX.

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TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	400	mV	29,30

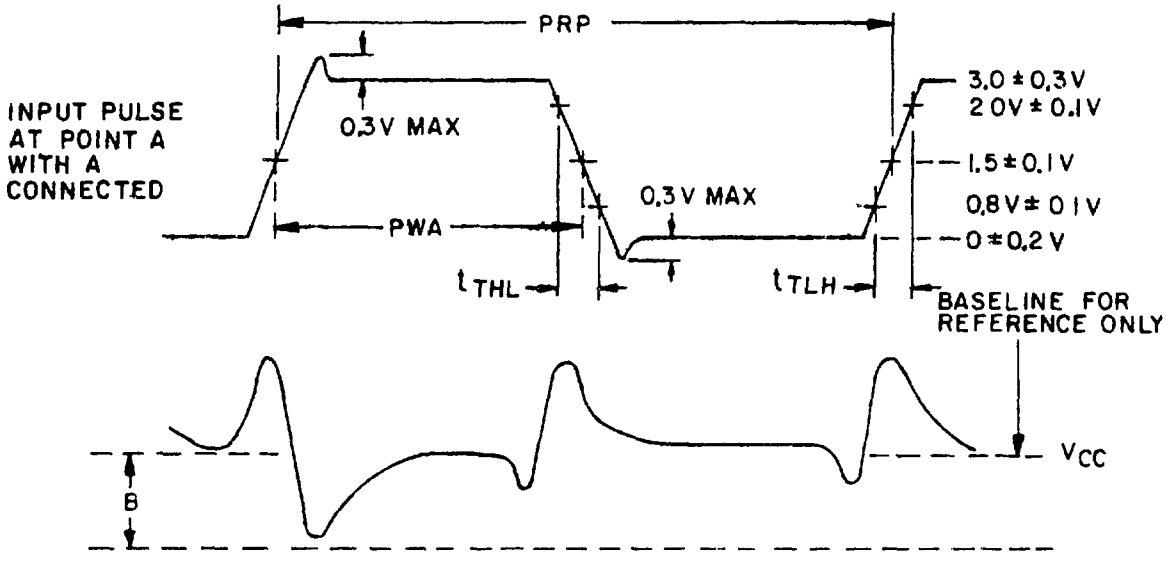
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 29. Power supply transient amplitude test circuit for table XX.

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NOTES:
1. Point A input parameters:
   PWA = 200 ±20 ns
   PRP = 400 ±40 ns
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns verified at each input.
2. B = Measurement from the baseline to the most negative excursion of the waveform.

FIGURE 30. Power supply transient amplitude test circuit waveforms for table XX.

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TABLE XXI. Input and output capacitance test.

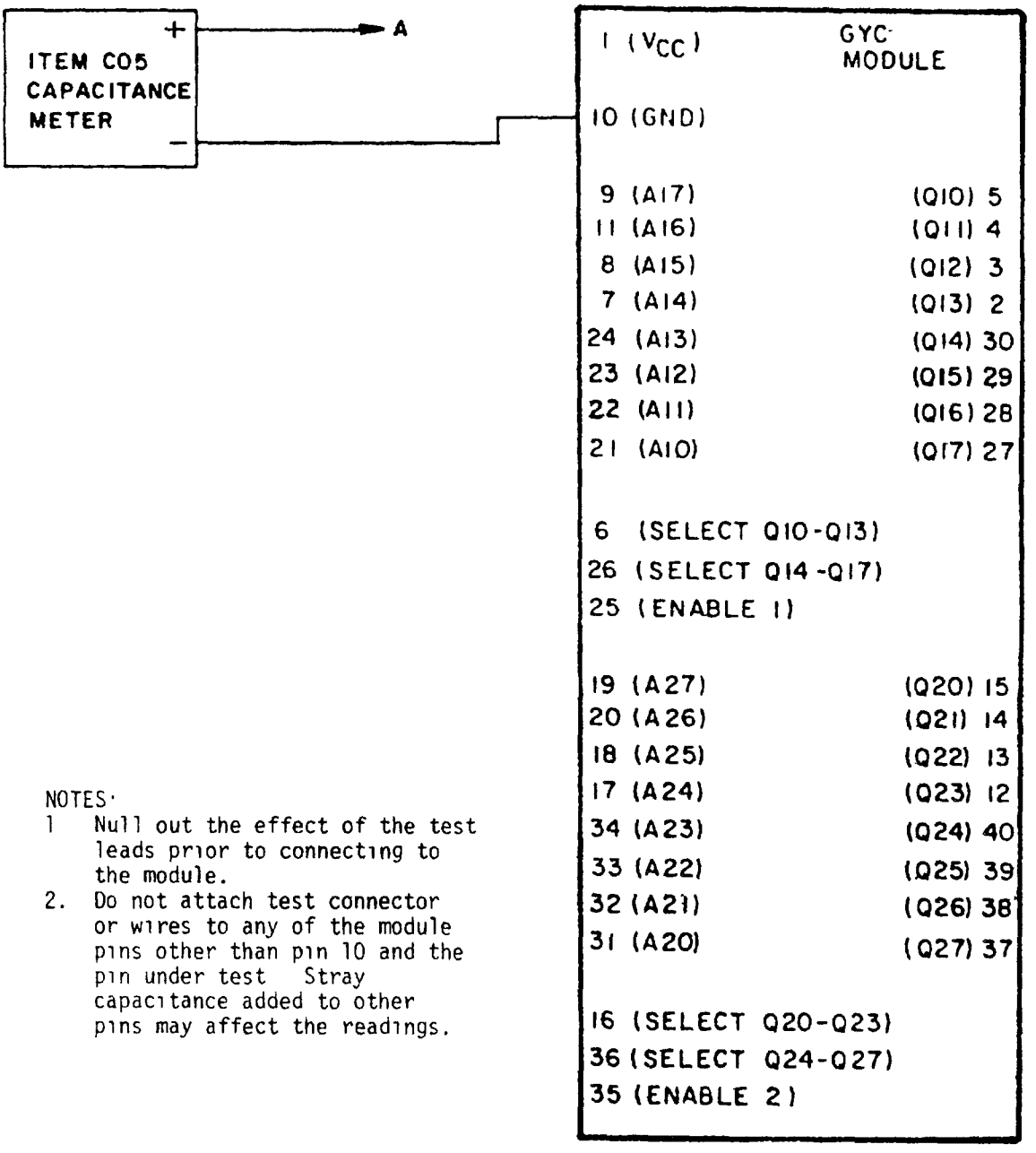
Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
Z1	A17 Input capacitance	9	2	30	pF	31
Z2	A16	11				
Z3	A15	8				
Z4	A14	7				
Z5	A13	24				
Z6	A12	23				
Z7	A11	22				
Z8	A10	21				
Z9	A27	19				
Z10	A26	20				
Z11	A25	18				
Z12	A24	17				
Z13	A23	34				
Z14	A22	33				
Z15	A21	32				
Z16	A20	31				
Z17	Enable 1	25				
Z18	Enable 2	35				
Z19	Select Q10-Q13	6		20		
Z20	Select Q20-Q23	16				
Z21	Select Q24-Q27	36				
Z22	Select Q14-Q17	26				

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TABLE XXI. Input and output capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
Z23	Q10 Output capacitance	5	2	15	pF	31
Z24	Q11	4				
Z25	Q12	3				
Z26	Q13	2				
Z27	Q14	30				
Z28	Q15	29				
Z29	Q16	28				
Z30	Q17	27				
Z31	Q24	40				
Z32	Q25	39				
Z33	Q26	38				
Z34	Q27	37				
Z35	Q20	15				
Z36	Q21	14				
Z37	Q22	13				
Z38	Q23	12				

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

*FIGURE 31. Input and output capacitance test circuit for table XXI.

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